

Agentic AI Supply Chain System

Complete Documentation — FMCG / Biscuit Sector

Literature Review · Market Data · Architecture · Pipelines · Gap Analysis · Scope

Document Version: 1.0

Prepared for: 2-Person Founding Team

Target Companies: Biskfarm, Britannia, ITC, Parle, Local FMCG Brands

Status: Research & Planning Phase

🍪 AGENTIC AI SUPPLY CHAIN SYSTEM

Complete Documentation — FMCG / Biscuit Sector

Literature Review · Market Data · Architecture · Pipelines · Gap Analysis · Scope

> ***Document Version:*** *1.0*
>
> ***Prepared for:*** *2-Person Founding Team*
>
> ***Target Companies:*** *Biskfarm, Britannia, ITC, Parle, Local FMCG Brands*
>
> ***Status:*** *Research & Planning Phase*

📋 TABLE OF CONTENTS

1. Executive Summary
2. Literature Review — Deep Dive
3. What We Have Currently — Knowledge State
4. Market Growth & Opportunity
5. Full System Architecture
6. Data Pipeline — Detailed Flow
7. Feature-Level Flow Diagrams
8. Agent Interaction Diagrams
9. Gap Analysis — What Exists vs What We Build
10. What We Need to Learn
11. Scope to Build & Improve
12. Phase-wise Execution Plan
13. Technology Stack — Deep Dive
14. Business Model & Monetisation
15. Risk Register
16. References & Sources

1. Executive Summary

This document is the **complete reference blueprint** for building an Agentic AI-powered supply chain intelligence platform for the biscuit and FMCG distribution sector in India.

The system addresses a **₹15,000 crore problem**: Indian FMCG distributors lose an estimated 8–15% of potential revenue annually due to stockouts, poor demand forecasting, over-stocking of slow movers, and missed festive demand peaks. No agentic, India-specific, end-to-end solution exists for this sector today.

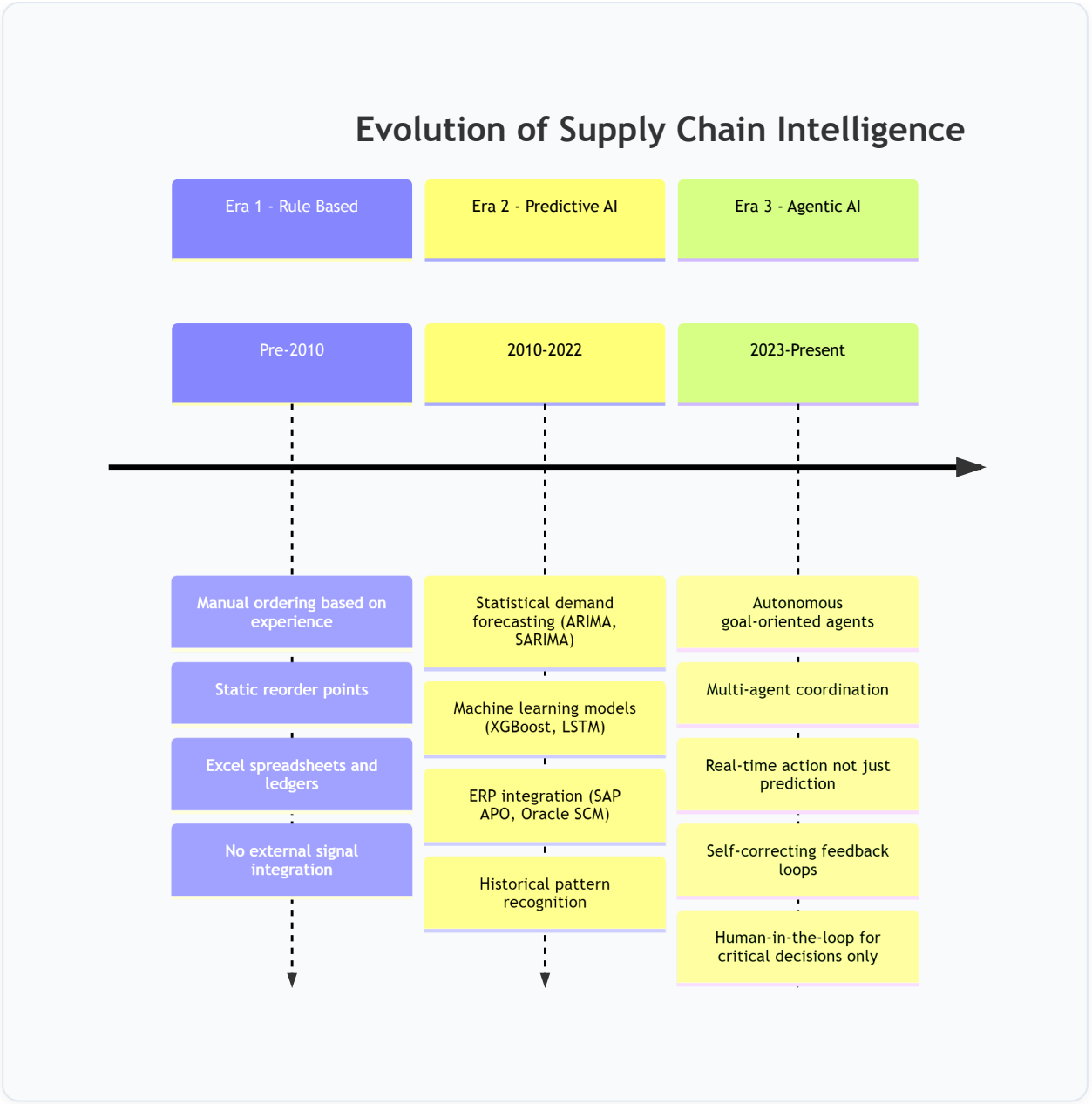
The 6 Core Capabilities We Are Building

```
┌──────────────────────────────────────────────────────────────┐
│  CAPABILITY 1  |  Area × SKU Demand Heatmap (which area needs what) |
│  CAPABILITY 2  |  Offer / Promotion Generator (AI-driven schemes)   |
│  CAPABILITY 3  |  Weekly Auto Stock Order (inventory intelligence)  |
│  CAPABILITY 4  |  Festive & Batch Demand Forecasting (spike alerts) |
│  CAPABILITY 5  |  Distribution Planning (weekly/bi-weekly beats)    |
│  CAPABILITY 6  |  Quick Commerce Supply Integration (dark stores)   |
└──────────────────────────────────────────────────────────────┘
```

2. Literature Review — Deep Dive

2.1 Supply Chain Intelligence — Historical Evolution

The field of supply chain management has evolved through three distinct paradigms:



The Core Difference: Predictive vs Agentic

DIMENSION	TRADITIONAL AI (PREDICTIVE)	AGENTIC AI
Output	"Demand will spike 22% next week"	Automatically raises PO, alerts distributor, adjusts beat plan
Decision	Human decides what to do	Agent decides and executes (with optional human approval)
Adaptability	Needs manual retraining	Self-updates from feedback loops
Error handling	Human intervention required	Critic agent reviews and re-plans
Speed	Days to respond	Minutes to respond
External signals	Limited inputs	Weather, events, social media, competitor data

---

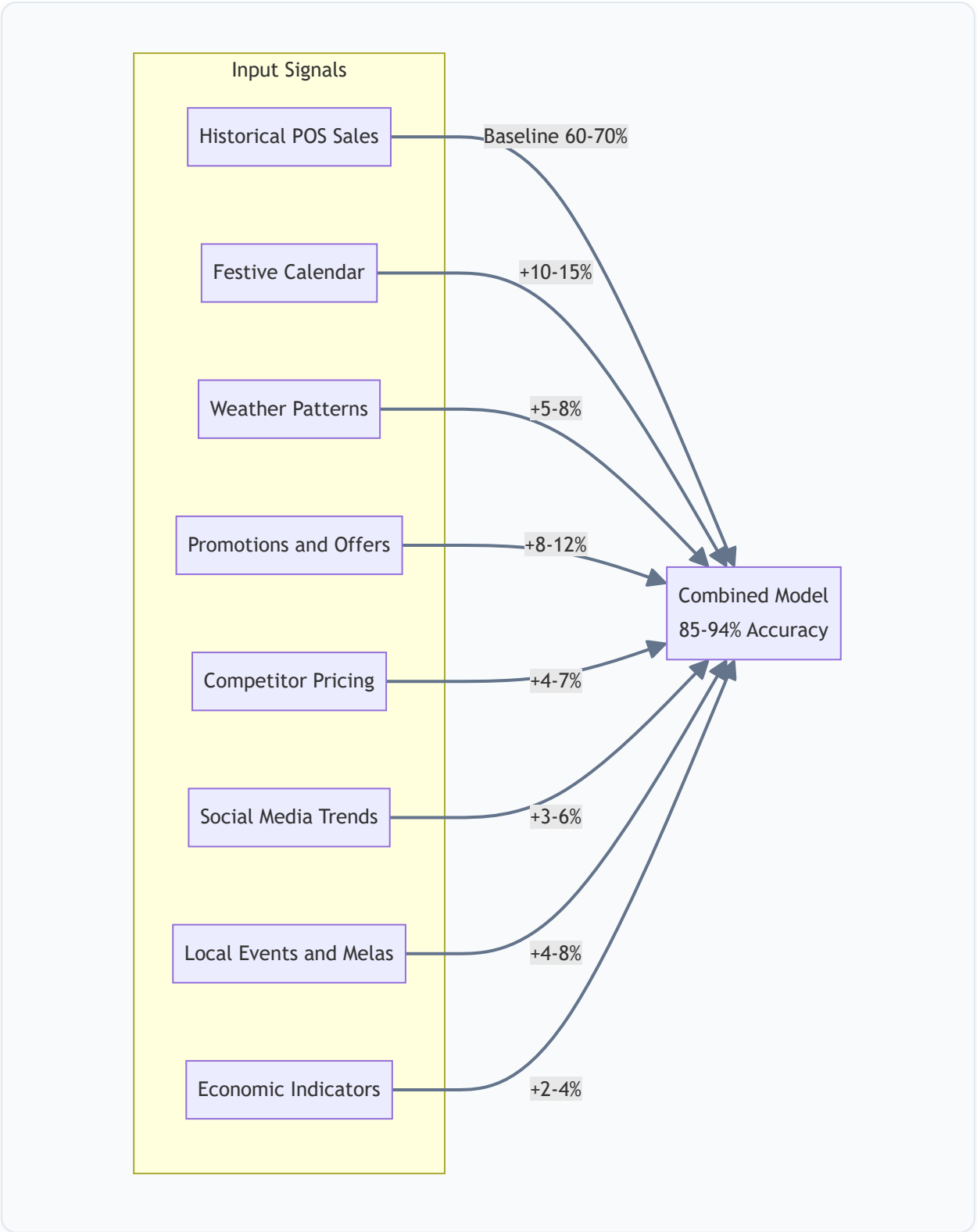
2.2 AI & ML in FMCG — Global Research

2.2.1 Demand Forecasting — State of the Art

Best-performing demand forecasting models for FMCG (as of 2024):

MODEL TYPE	ACCURACY RANGE	BEST FOR	LIMITATION
SARIMA	60–72%	Stable, low-volatility SKUs	Cannot handle promotions or events
XGBoost / LightGBM	75–85%	Mixed signals, tabular data	Needs large historical dataset
Facebook Prophet	72–82%	Seasonal patterns, holidays	Poor on short-history SKUs
LSTM Neural Networks	78–88%	Complex time-series patterns	Computationally expensive
Ensemble (hybrid)	85–92%	Production systems	More complex to maintain
LLM + RAG (emerging)	88–94%	Multi-signal integration	Needs careful prompt engineering

2.2.2 Key Demand Signals & Their Proven Impact



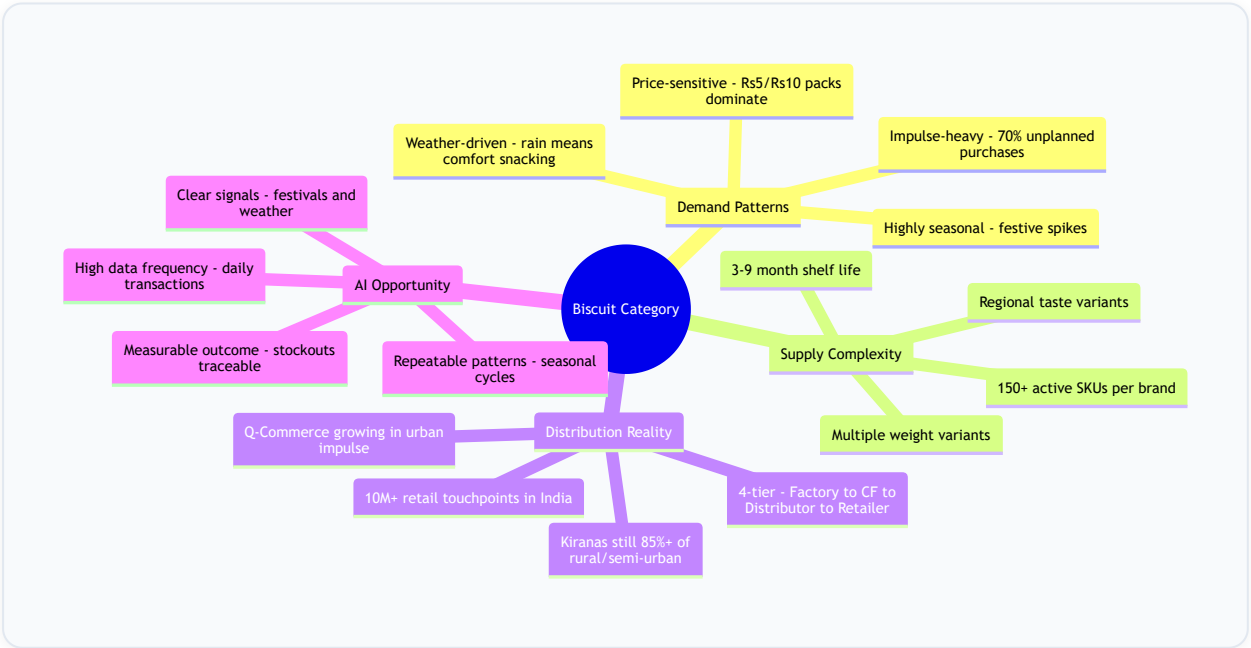
2.2.3 ROI Evidence from FMCG Implementations

Published industry case studies (IBM, Deloitte, McKinsey, 2023–2024):

KPI	IMPROVEMENT REPORTED	SOURCE
Stockout rate reduction	15–30%	Deloitte FMCG Supply Chain Report 2024
Forecast error (MAPE) improvement	20–50%	IBM Institute for Business Value 2024
Inventory holding cost reduction	25–35%	McKinsey Operations Practice 2023
Working capital released	18–28%	Accenture Supply Chain Survey 2024
Order processing time	Days → Minutes	Kanerika AI Supply Chain Study 2024
Revenue growth vs. non-AI peers	5.8× faster	IBM IBV High Performer Analysis 2024

2.3 Biscuit Industry — Category-Specific Research

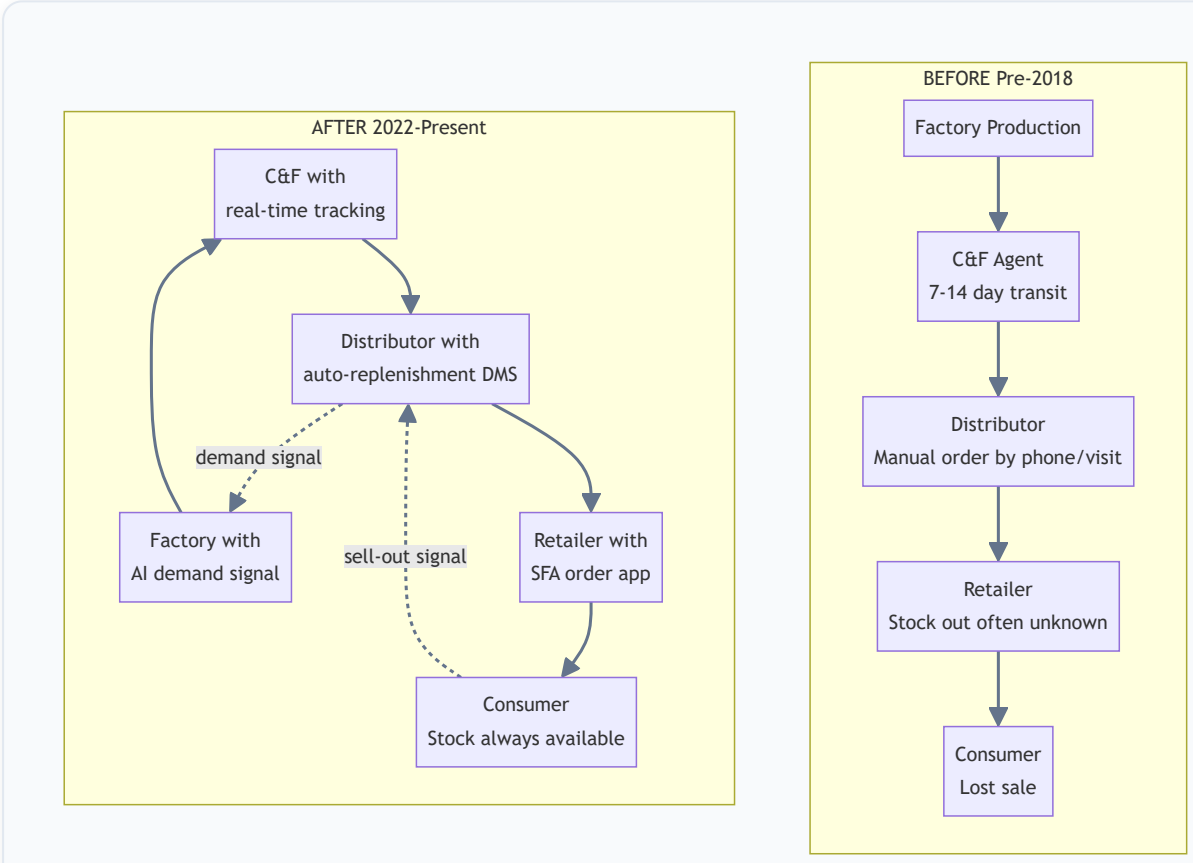
2.3.1 Why Biscuits Are a Uniquely Interesting Category



2.3.2 Regional Preference Map

REGION	DOMINANT BISCUIT TYPE	KEY BRANDS	OCCASION
North India (UP, Bihar, Punjab)	Glucose, cream biscuits	Parle-G, Britannia Tiger	Daily consumption, tea
South India (TN, Kerala, Andhra)	Digestive, whole wheat	Sunfeast, Horlicks	Health-conscious, gifting
West India (Gujarat, Maharashtra)	Cream, wafers	Good Day, Bourbon	Premium, urban
East India (Bengal, Odisha)	Sweet, coconut	Biskfarm, local brands	Festive, pooja
Tier-2/3 Cities (all regions)	Economy packs (₹5–₹10)	Parle, local	Volume, affordability

2.3.3 Britannia — The Most Studied Case in India



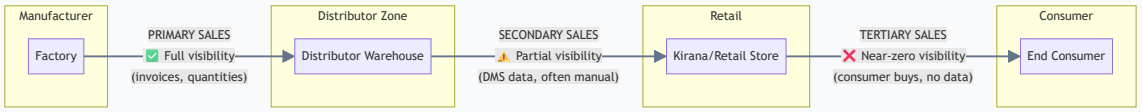
Measured outcomes after Britannia's transformation:

- Lead time reduced by **30–40%**
- Inventory holding cut by **~25%**

- Stockout incidents down **~20%** in digitized zones

2.4 India FMCG Distribution — Unique Context

2.4.1 The Visibility Gap Problem



> ***The Visibility Gap*** *= The manufacturer only reliably sees primary sales. They are effectively* ***blind*** *to real consumer demand. This causes the* ***bullwhip effect*** *— demand distortion that amplifies upstream.*

2.4.2 India Distribution Scale — Numbers

METRIC	FIGURE
Total retail outlets in India	~12–14 million
Kirana stores (traditional)	~10–12 million (85%+)
Modern trade outlets	~50,000–60,000
Q-commerce dark stores (2024)	~4,000–5,000 (growing fast)
Typical biscuit company distributors	2,000–8,000 per company
Retailers per distributor	200–1,500
Distributor tech adoption (DMS)	~35–40%
Distributor using Excel/manual	~55–60%

2.4.3 GST as the Hidden Data Gold Mine

Post-2017 GST implementation has created an unprecedented digital paper trail. Every invoice is digitally filed. With distributor consent, this provides **granular secondary sales data** that was impossible to access pre-2017 — a uniquely Indian advantage.

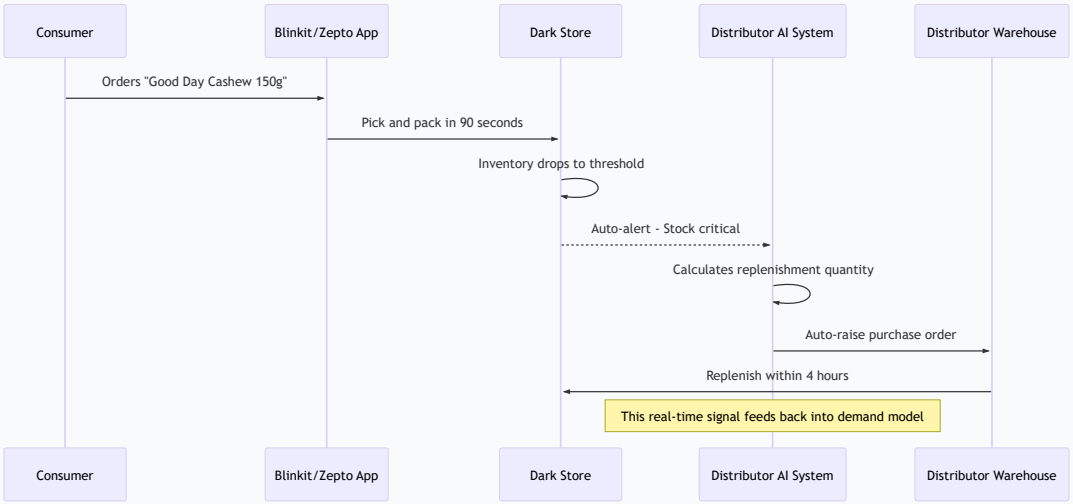
2.5 Quick Commerce — The New Distribution Layer

2.5.1 Q-Commerce Market Growth (India)

YEAR	GMV (USD)	KEY MILESTONE
2022	$1.5B	Blinkit, Zepto, Swiggy Instamart scaling
2024	$6–7B	Q-comm = 2/3rd of all e-grocery orders
2027	~$22B	Expansion to Tier-2 cities
2030	$40–50B	10% of total branded retail sales

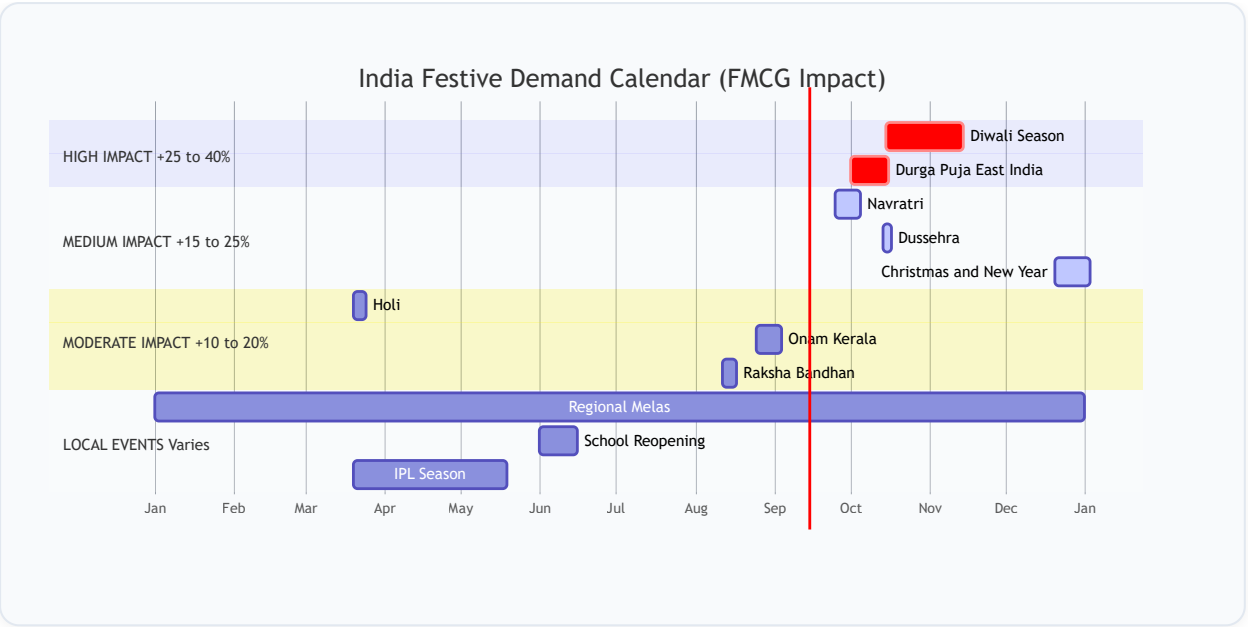
2.5.2 Why Q-Commerce Changes Biscuit Supply Chains

Q-commerce reveals demand **in real-time** (vs. kirana which hides it for 3–7 days):



2.6 Festive Demand & Promotional Planning Research

2.6.1 India's Festive Calendar — Demand Impact



2.6.2 Promotional Effectiveness Research

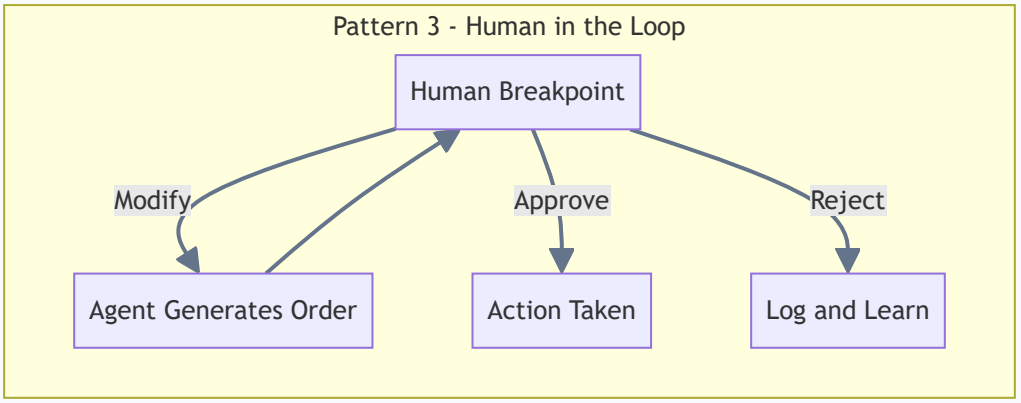
METRIC	MANUAL PLANNING	AI-DRIVEN PLANNING	IMPROVEMENT
Promotion planning lead time	4–6 weeks	1–2 weeks	60–70% faster
Stockout during promo period	18–25% of SKUs	5–8% of SKUs	~70% reduction
Excess stock after promo ends	12–20% unsold	3–7% unsold	~65% reduction
Distributor scheme claim accuracy	65–75%	90–95%	~25% better
ROI per promotional rupee	₹1.8–2.2	₹2.8–3.5	40–60% better

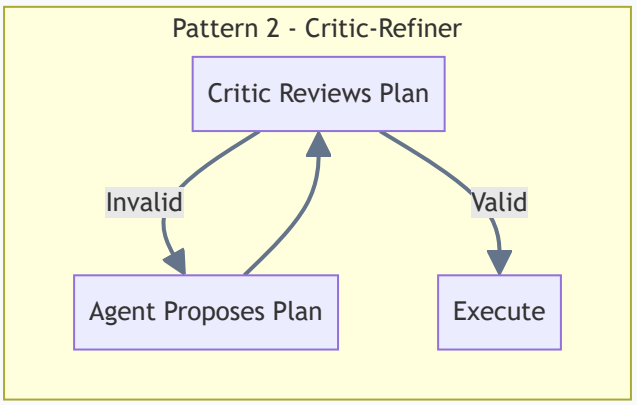
2.7 Multi-Agent AI Systems — Architecture Research

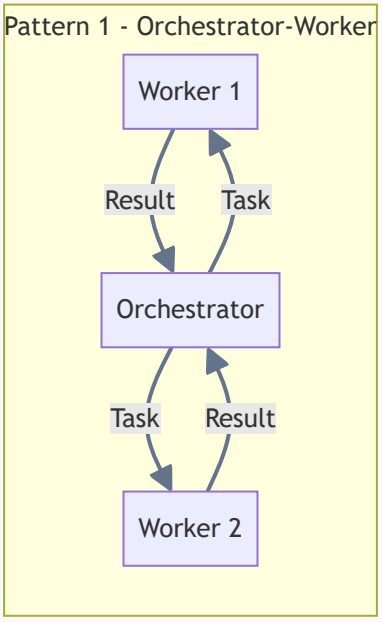
2.7.1 Framework Comparison

FEATURE	LANGGRAPH	CREWAI
Philosophy	Graph-based state machine	Role-based team simulation
Best for	Complex conditional logic, audit trails	Quick collaborative workflows
State management	Explicit, persistent, customizable	Simplified via task outputs
Human-in-the-loop	Native support (breakpoints)	Plugin-based
Production readiness	Enterprise-grade	Excellent for operational scaling
Our recommendation	Phase 3+ (complex agents)	Phase 1–2 (fast prototyping)

2.7.2 Design Patterns We Will Implement

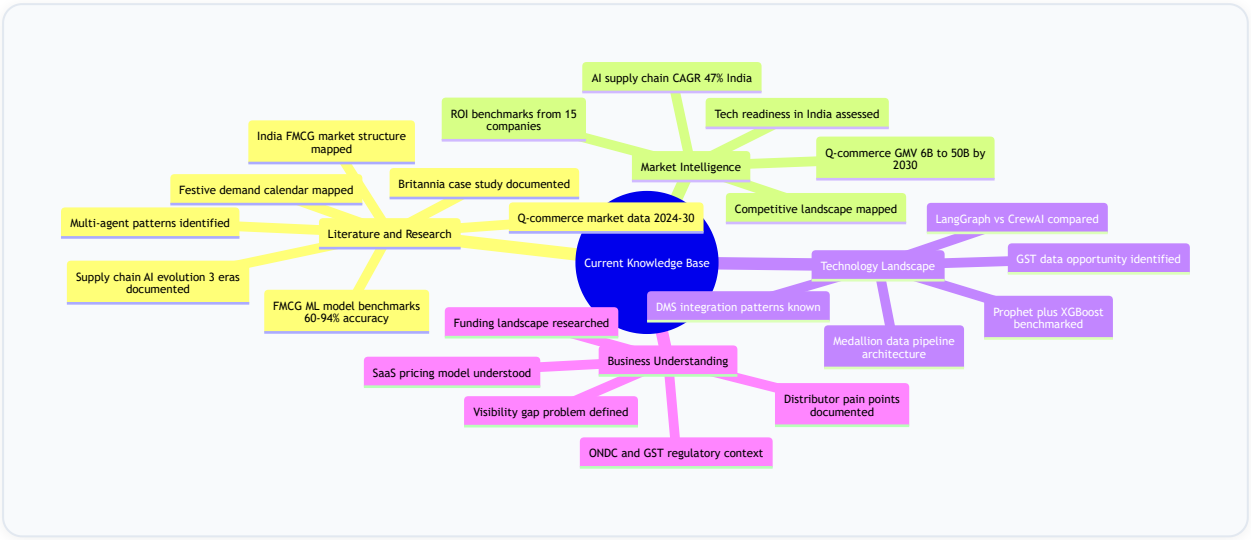

Pattern 3 - Human in the Loop
Human Breakpoint
Modify
Approve
Reject
Agent Generates Order
Action Taken
Log and Learn


Pattern 2 - Critic-Refiner
Critic Reviews Plan
Invalid
Valid
Agent Proposes Plan
Execute


Pattern 1 - Orchestrator-Worker
Worker 1
Result
Task
Orchestrator
Task
Result
Worker 2

3. What We Have Currently — Knowledge State

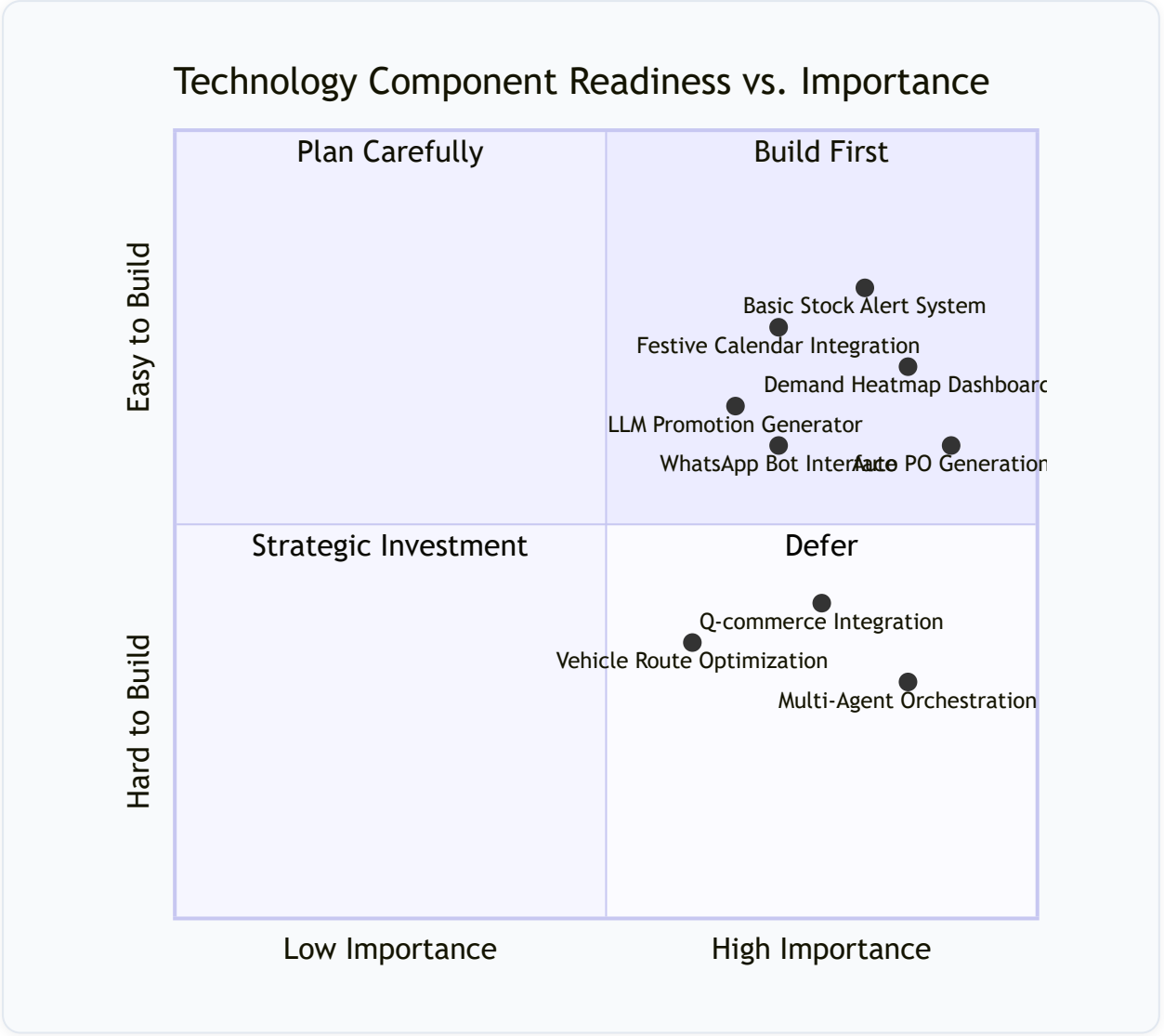
3.1 Research Knowledge Gathered



3.2 Data We Need to Gather (Before Building)

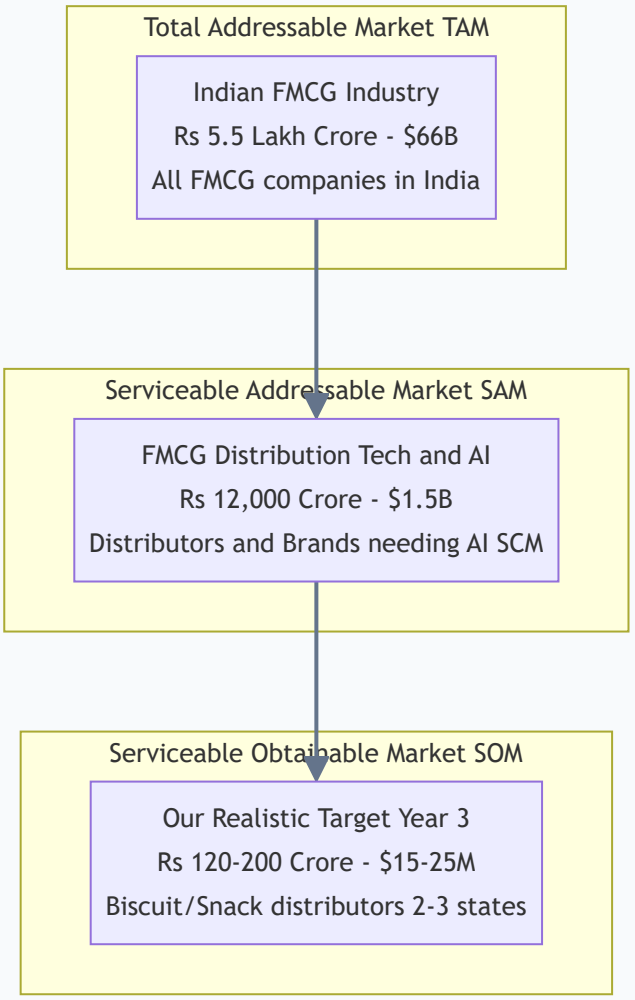
DATA TYPE	SOURCE	STATUS	PRIORITY
Historical sales data (1 distributor, 1–2 years)	Pilot distributor	❌ Not acquired	🔴 Critical
SKU master list (products, variants, prices)	Target FMCG company	❌ Not acquired	🔴 Critical
Retailer database (area-wise)	Field survey / distributor	❌ Not acquired	🔴 Critical
Festive calendar API	Public / custom built	⚠️ Partially available	🟡 High
Regional weather API	OpenWeatherMap, IMD	✅ Available (API)	🟢 Ready
Q-commerce dark store locations	Public data / partnerships	⚠️ Partially available	🟡 High
Competitor pricing feeds	Web scraping / reports	❌ Not built	🟠 Medium
Local event / mela data	Government portals	❌ Not built	🟠 Medium

3.3 Technology Readiness Assessment



4. Market Growth & Opportunity

4.1 Target Market Sizing — India



4.2 Global AI Supply Chain Market

MARKET	2023	2028	CAGR
AI in Global Supply Chain	$6.2B	$28.3B	35.4%
AI in Indian FMCG Supply Chain	$0.4B	$2.8B	**47.0%**
FMCG Distribution Tech India	$1.1B	$4.5B	32.0%
Q-Commerce India	$6.5B	~$35B	~40.0%
Last-Mile Delivery India	$7.5B	$20B+	12–13%

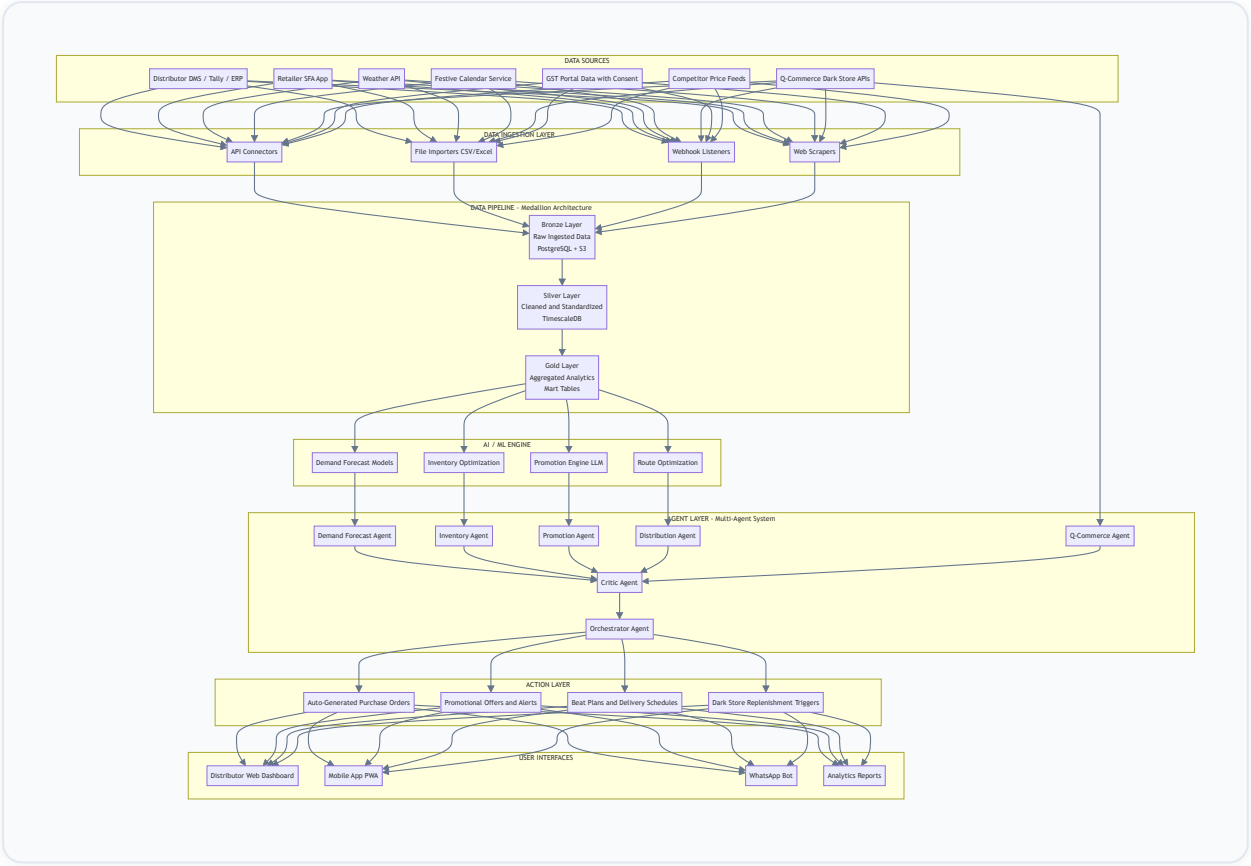
4.3 Distributor Pain Points — Financial Impact

A typical mid-size FMCG distributor with ₹1 Crore monthly turnover loses:

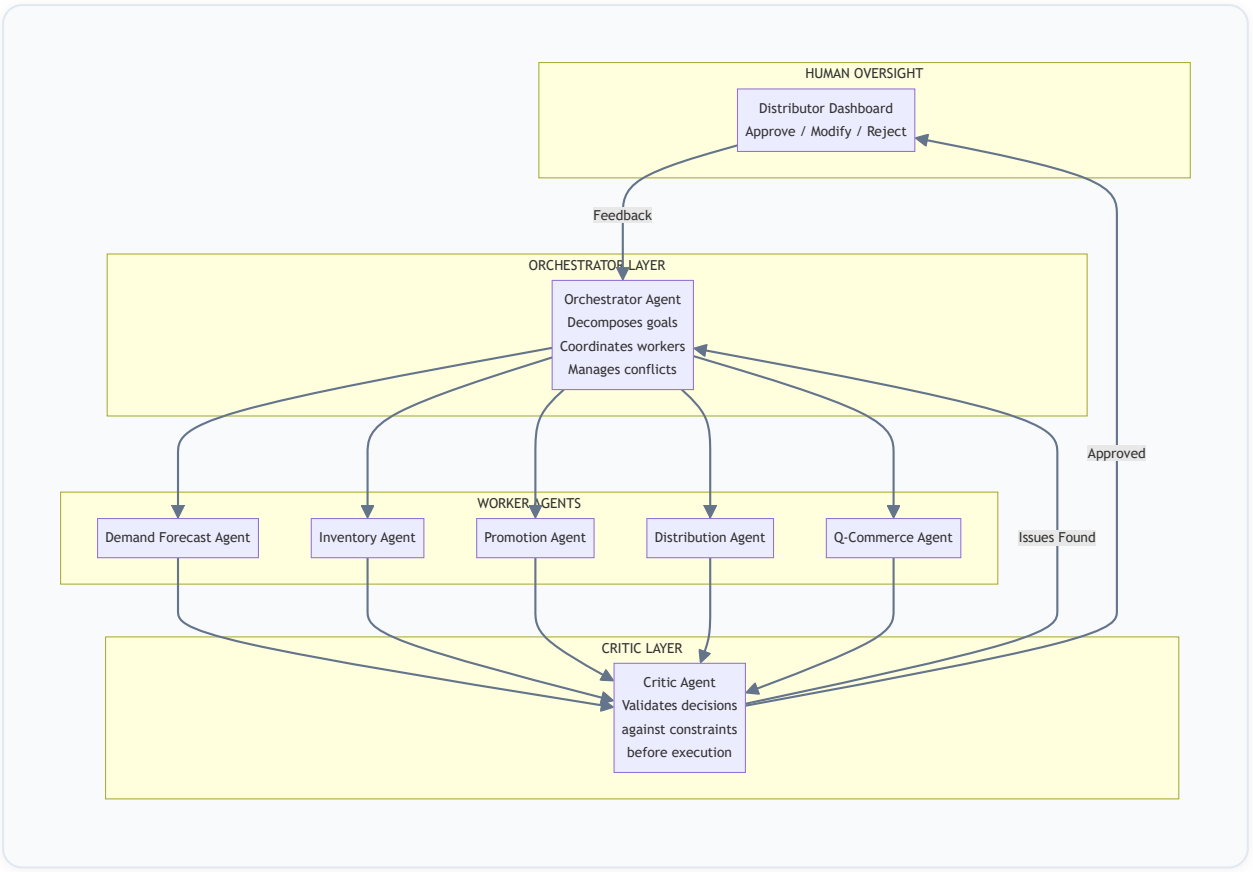
PAIN POINT	MONTHLY LOSS ESTIMATE
Stockouts (lost sales)	₹4–8 Lakhs
Over-stocking (carrying cost + expiry write-offs)	₹2–5 Lakhs
Manual order processing (labour cost)	₹50K–1.5 Lakhs
Missed promotional scheme claims	₹1–3 Lakhs
Sub-optimal delivery routes	₹75K–2 Lakhs
Total Monthly Pain	**₹8–19 Lakhs**
Annual Pain	**₹96 Lakhs – ₹2.3 Crores**

5. Full System Architecture

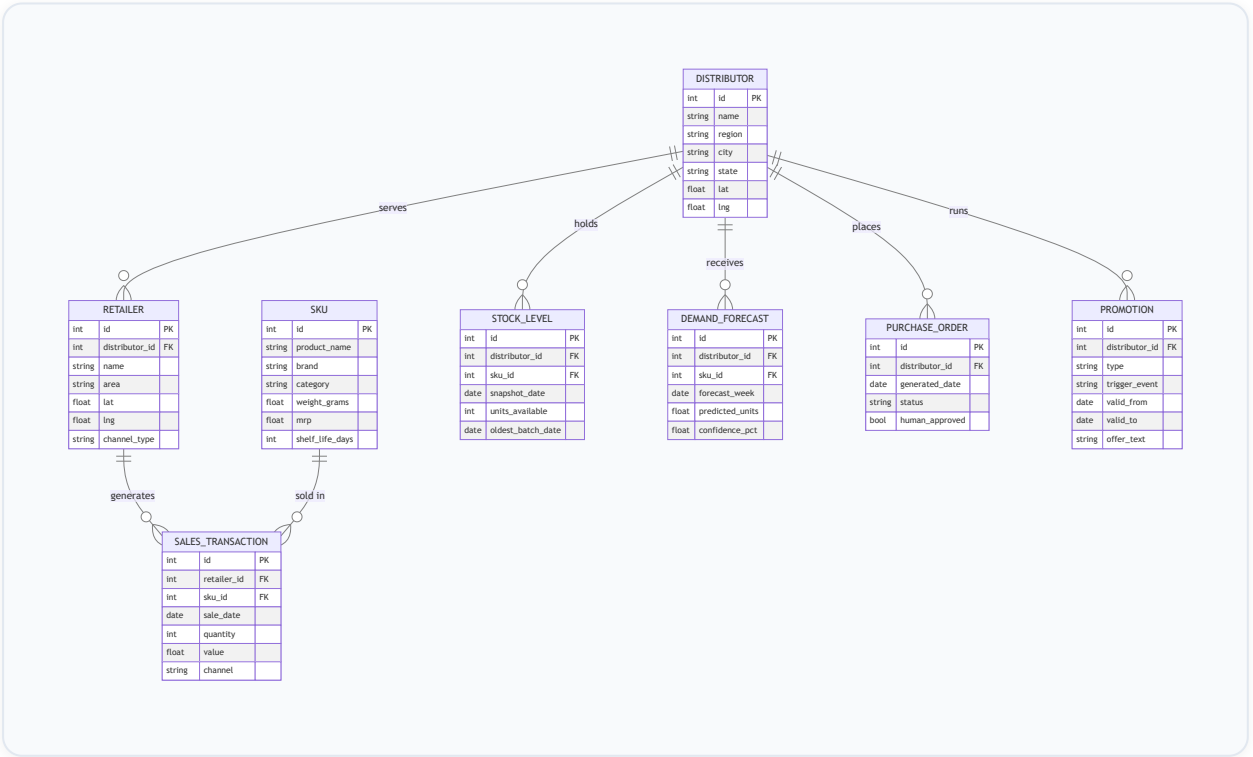
5.1 High-Level Architecture Overview



5.2 Multi-Agent Orchestration Detail

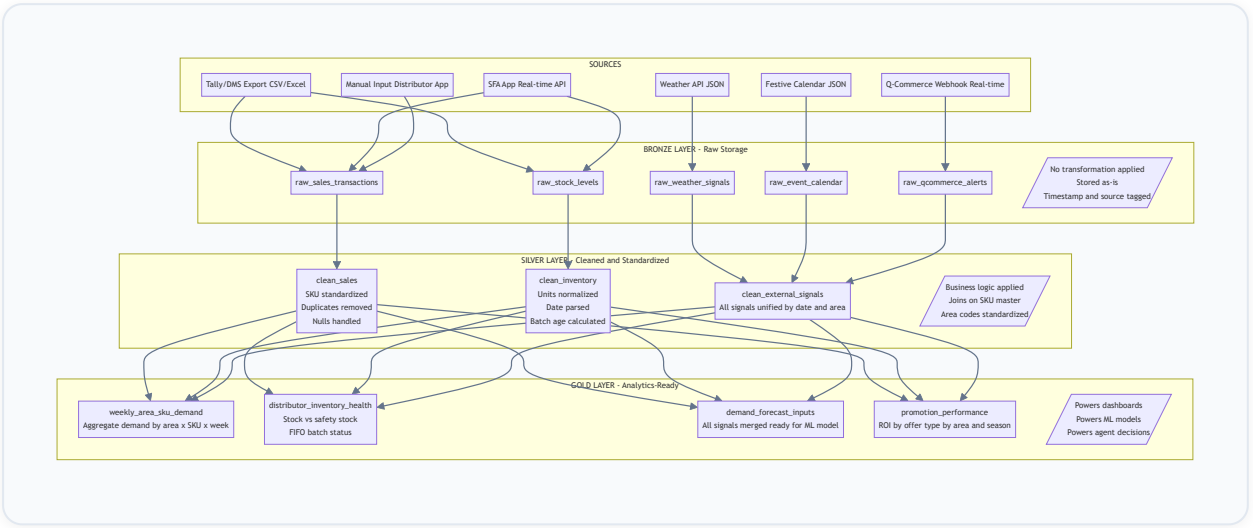


5.3 Database Schema (Conceptual)

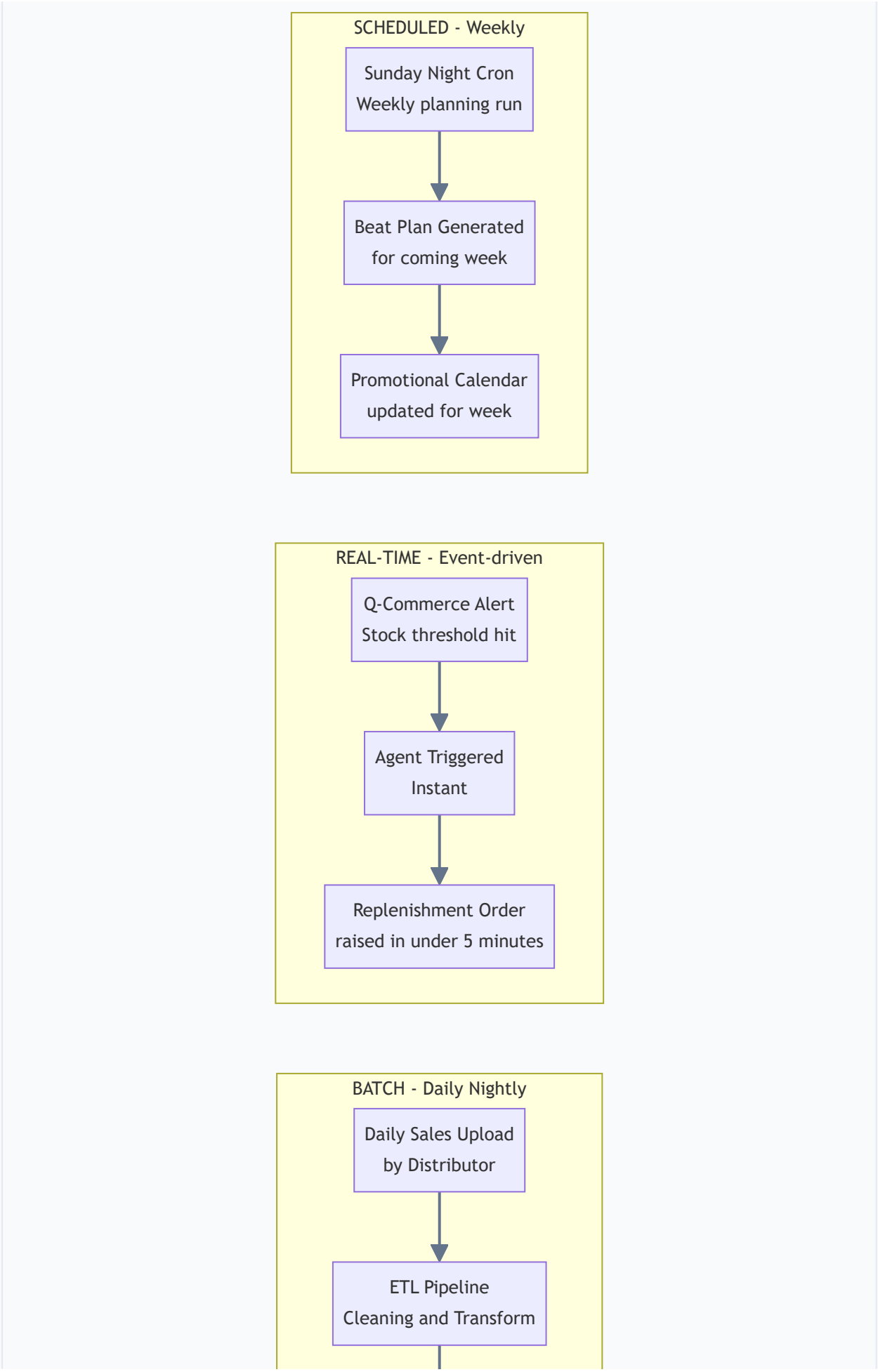


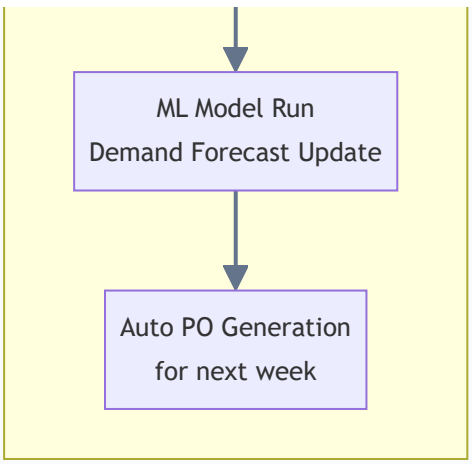
6. Data Pipeline — Detailed Flow

6.1 The Medallion Architecture



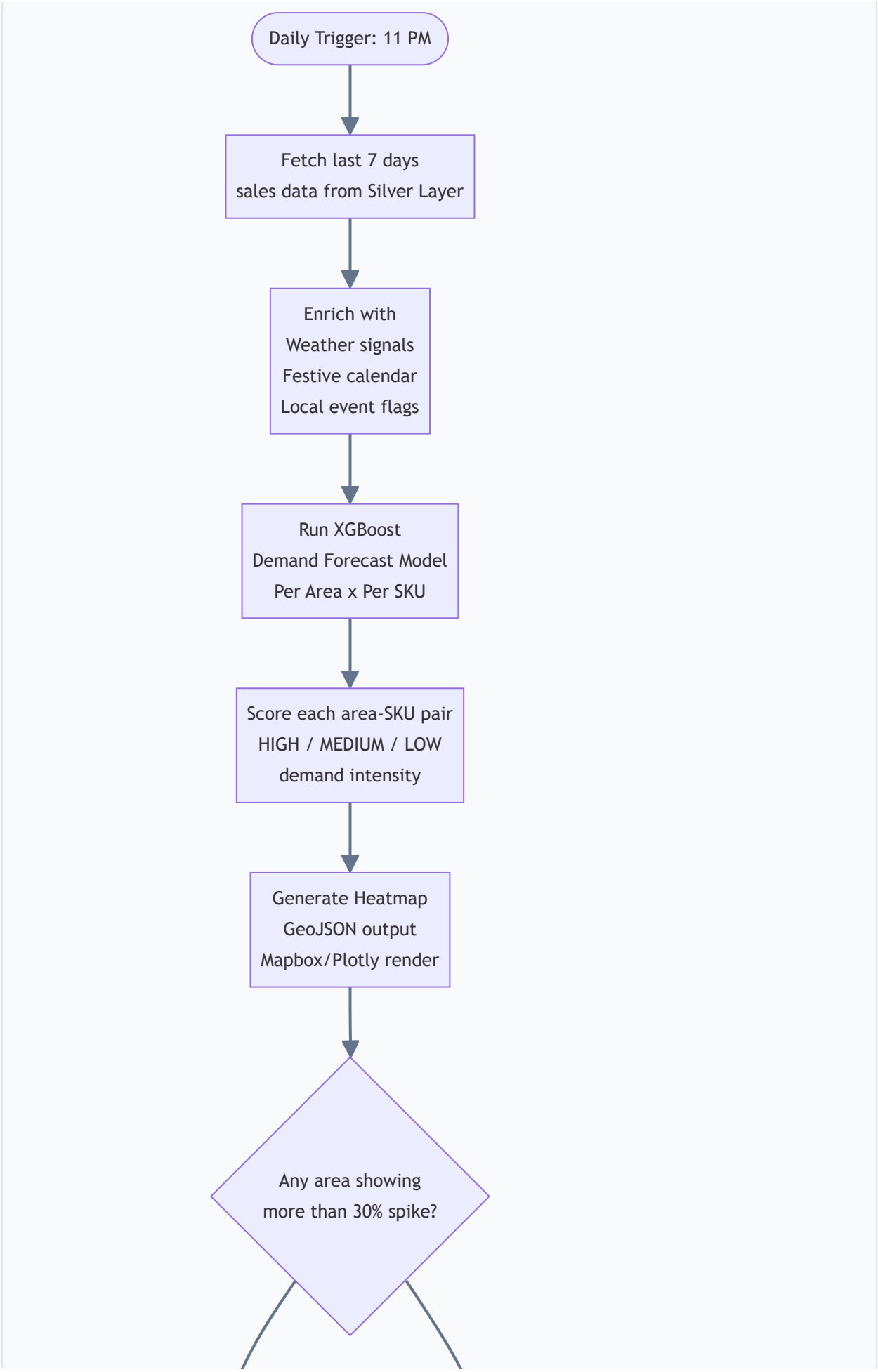
6.2 Real-time vs. Batch Processing

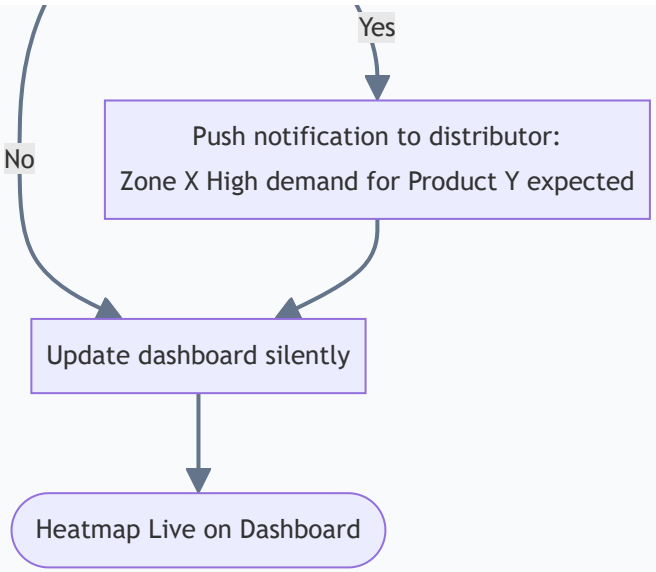

SCHEDULED - Weekly
Sunday Night Cron
Weekly planning run
Beat Plan Generated
for coming week
Promotional Calendar
updated for week
REAL-TIME - Event-driven
Q-Commerce Alert
Stock threshold hit
Agent Triggered
Instant
Replenishment Order
raised in under 5 minutes
BATCH - Daily Nightly
Daily Sales Upload
by Distributor
ETL Pipeline
Cleaning and Transform


ML Model Run
Demand Forecast Update
Auto PO Generation
for next week

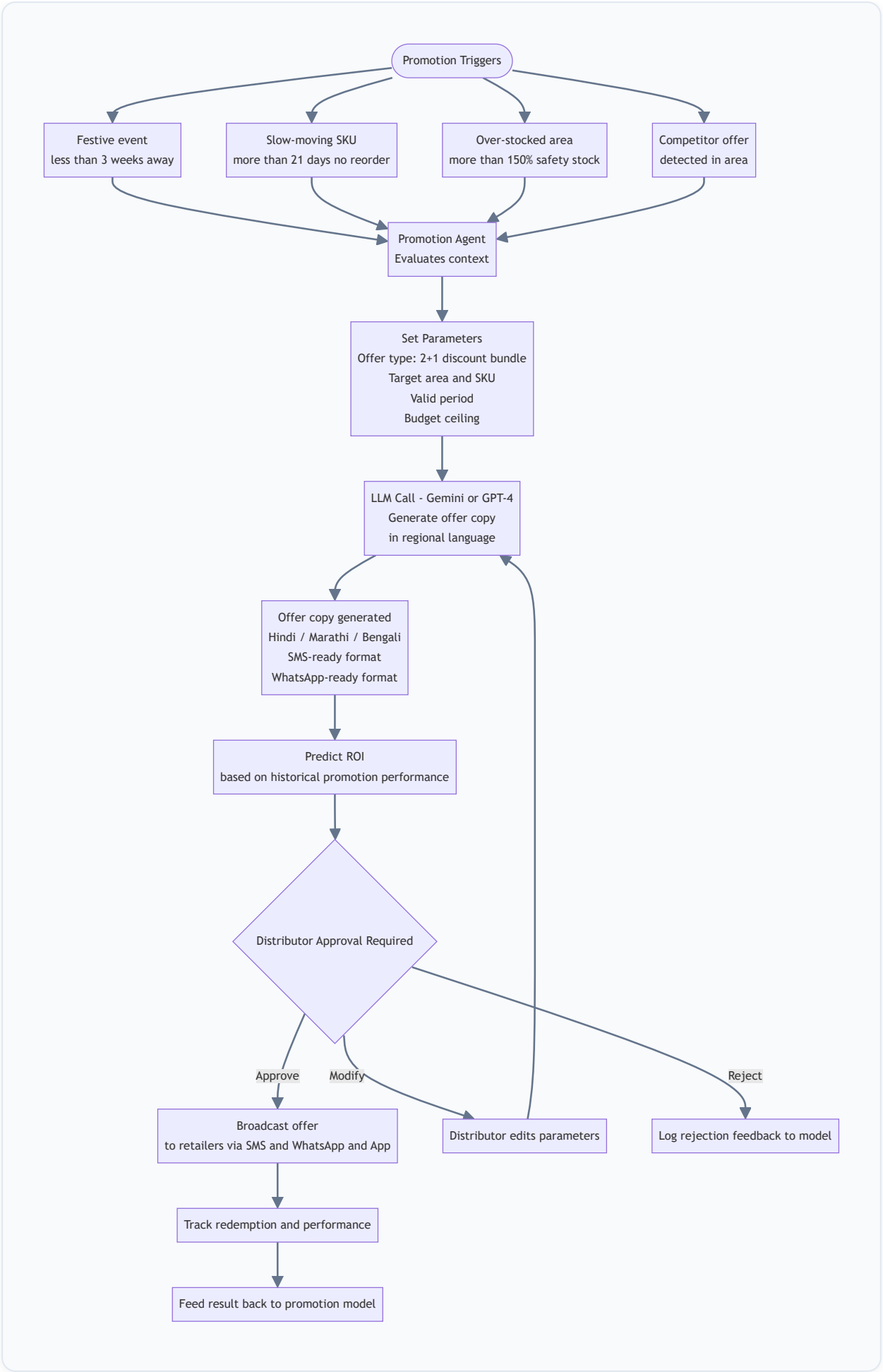
7. Feature-Level Flow Diagrams

7.1 Feature 1: Area × SKU Demand Heatmap

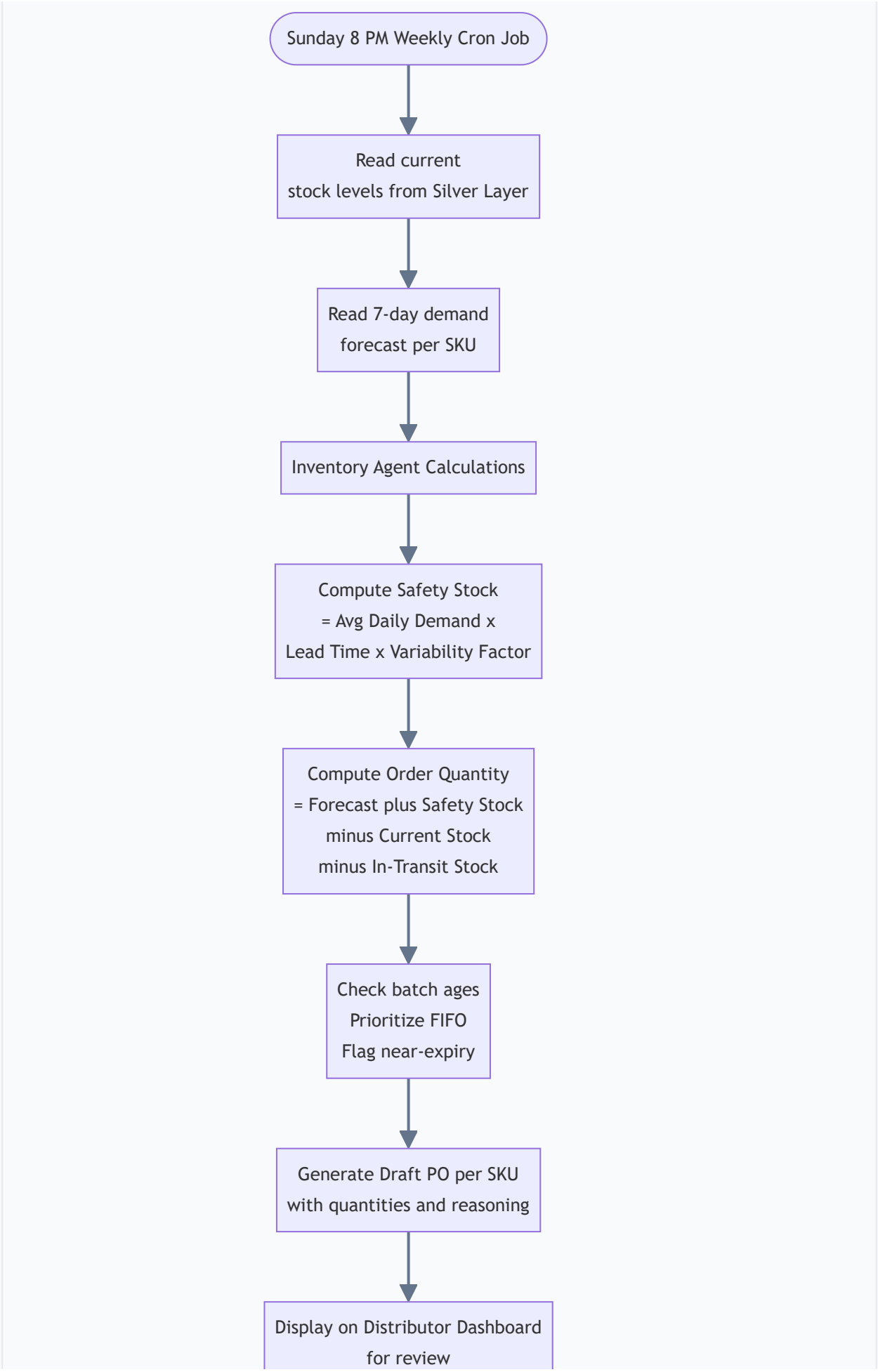

Daily Trigger: 11 PM
Fetch last 7 days
sales data from Silver Layer
Enrich with
Weather signals
Festive calendar
Local event flags
Run XGBoost
Demand Forecast Model
Per Area x Per SKU
Score each area-SKU pair
HIGH / MEDIUM / LOW
demand intensity
Generate Heatmap
GeoJSON output
Mapbox/Plotly render
Any area showing
more than 30% spike?

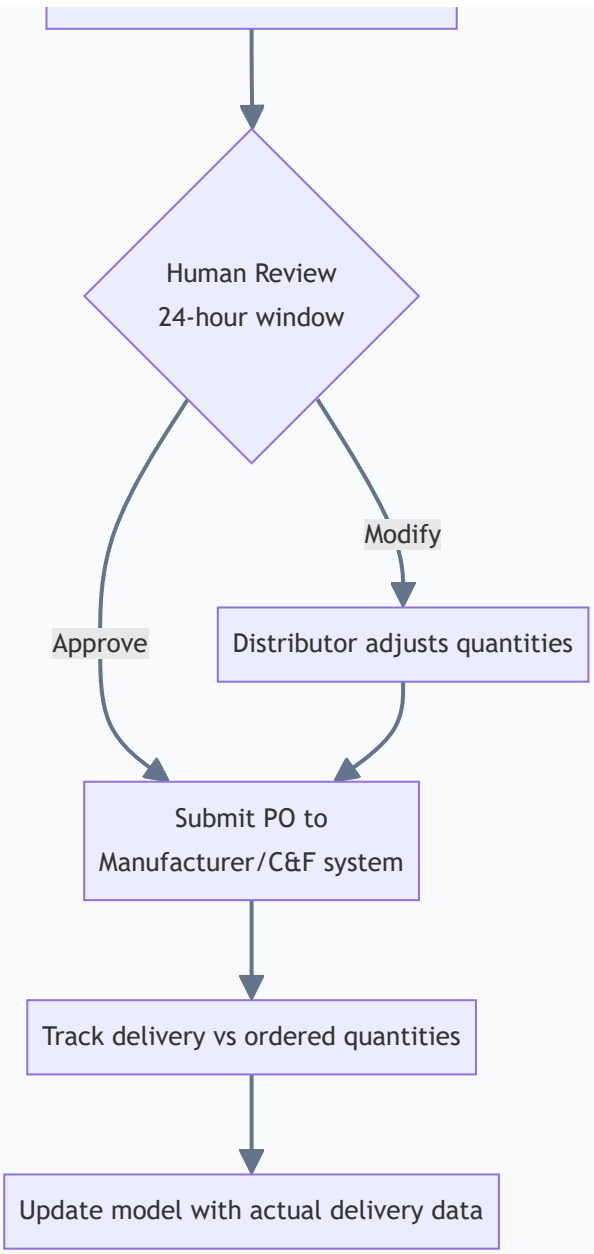

Yes
No
Push notification to distributor:
Zone X High demand for Product Y expected
Update dashboard silently
Heatmap Live on Dashboard

7.2 Feature 2: Offer & Promotion Generator

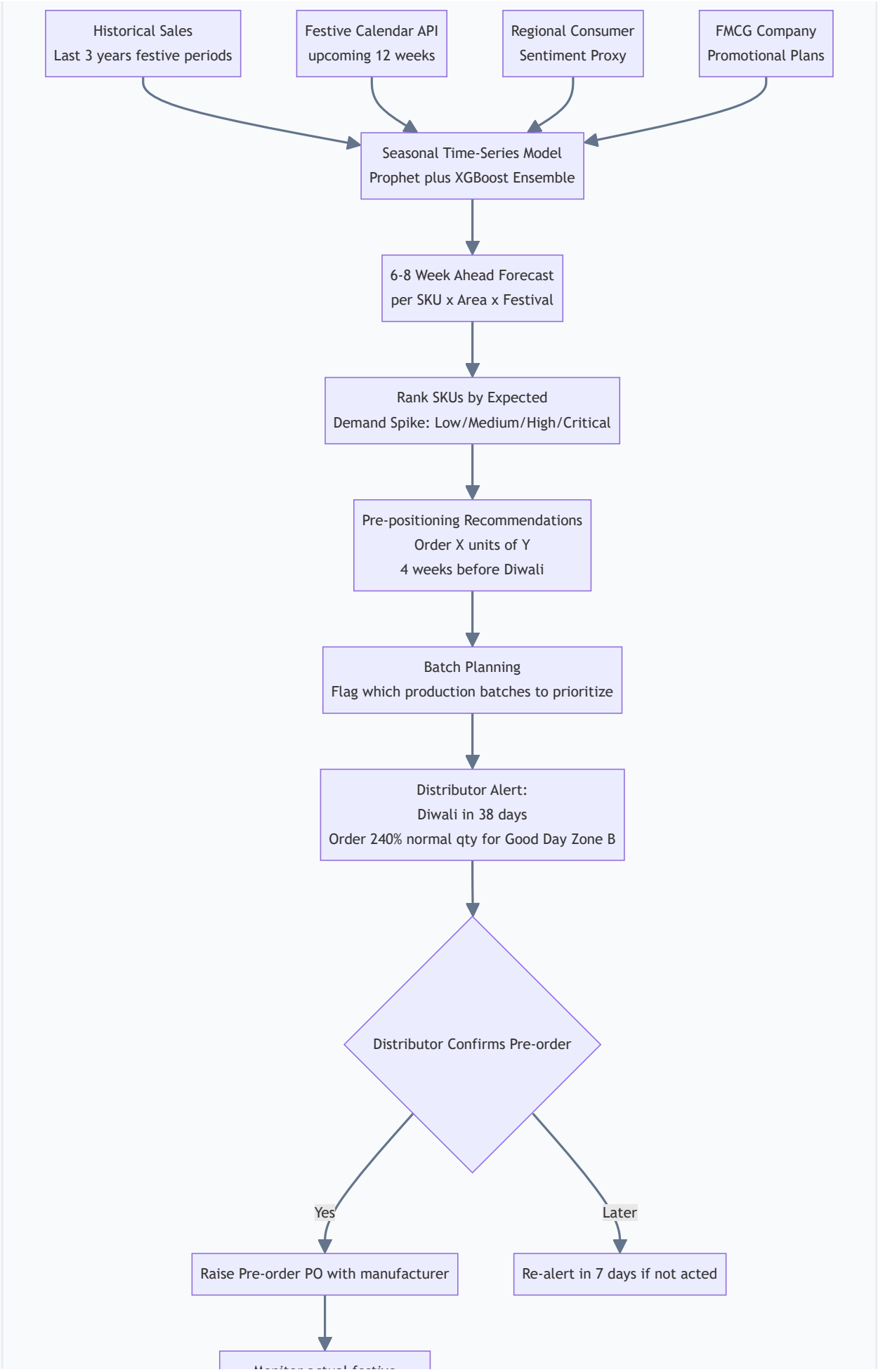

Promotion Triggers
Festive event
less than 3 weeks away
Slow-moving SKU
more than 21 days no reorder
Over-stocked area
more than 150% safety stock
Competitor offer
detected in area
Promotion Agent
Evaluates context
Set Parameters
Offer type: 2+1 discount bundle
Target area and SKU
Valid period
Budget ceiling
LLM Call - Gemini or GPT-4
Generate offer copy
in regional language
Offer copy generated
Hindi / Marathi / Bengali
SMS-ready format
WhatsApp-ready format
Predict ROI
based on historical promotion performance
Distributor Approval Required
Approve
Modify
Reject
Broadcast offer
to retailers via SMS and WhatsApp and App
Distributor edits parameters
Log rejection feedback to model
Track redemption and performance
Feed result back to promotion model

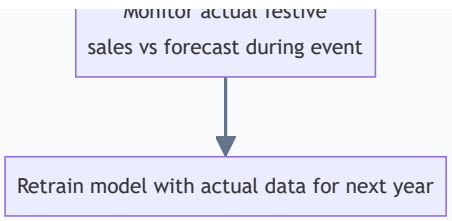
7.3 Feature 3: Weekly Stock Order Generation


Sunday 8 PM Weekly Cron Job
Read current stock levels from Silver Layer
Read 7-day demand forecast per SKU
Inventory Agent Calculations
Compute Safety Stock = Avg Daily Demand x Lead Time x Variability Factor
Compute Order Quantity = Forecast plus Safety Stock minus Current Stock minus In-Transit Stock
Check batch ages Prioritize FIFO Flag near-expiry
Generate Draft PO per SKU with quantities and reasoning
Display on Distributor Dashboard for review

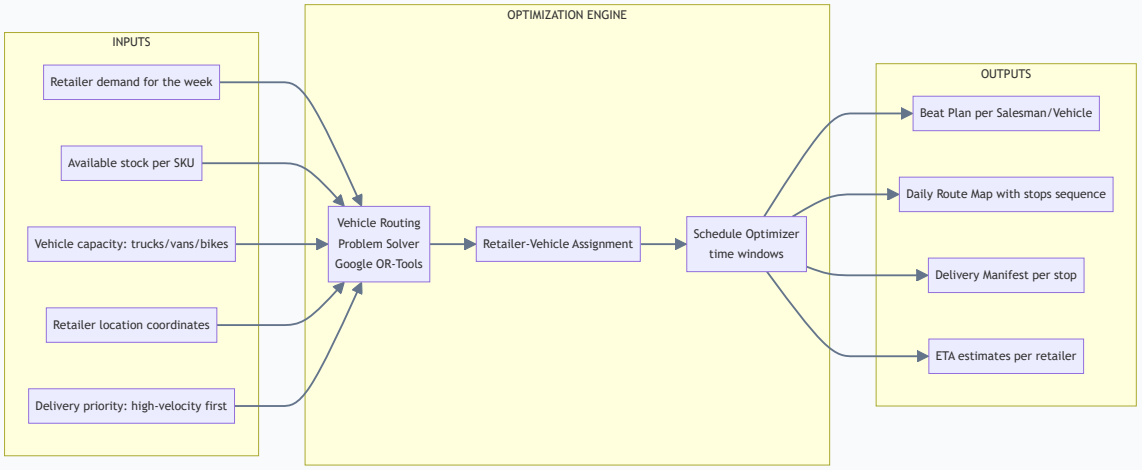

Human Review
24-hour window
Modify
Approve
Distributor adjusts quantities
Submit PO to
Manufacturer/C&F system
Track delivery vs ordered quantities
Update model with actual delivery data

7.4 Feature 4: Festive Batch Demand Forecasting

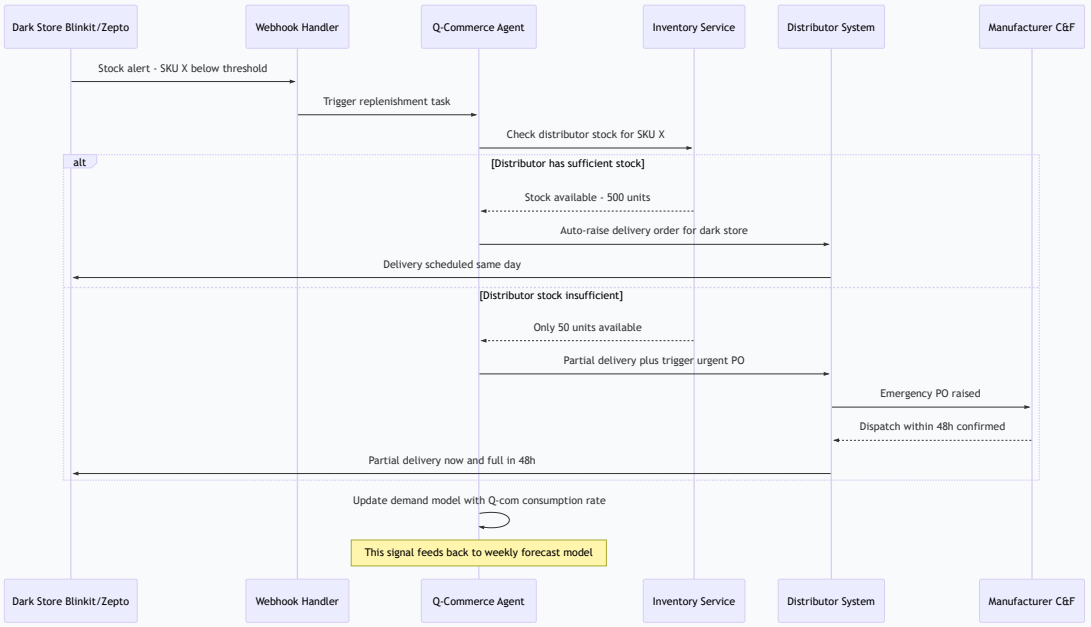

Historical Sales
Last 3 years festive periods
Festive Calendar API
upcoming 12 weeks
Regional Consumer
Sentiment Proxy
FMCG Company
Promotional Plans
Seasonal Time-Series Model
Prophet plus XGBoost Ensemble
6-8 Week Ahead Forecast
per SKU x Area x Festival
Rank SKUs by Expected
Demand Spike: Low/Medium/High/Critical
Pre-positioning Recommendations
Order X units of Y
4 weeks before Diwali
Batch Planning
Flag which production batches to prioritize
Distributor Alert:
Diwali in 38 days
Order 240% normal qty for Good Day Zone B
Distributor Confirms Pre-order
Yes
Later
Raise Pre-order PO with manufacturer
Re-alert in 7 days if not acted



7.5 Feature 5: Distribution Planning

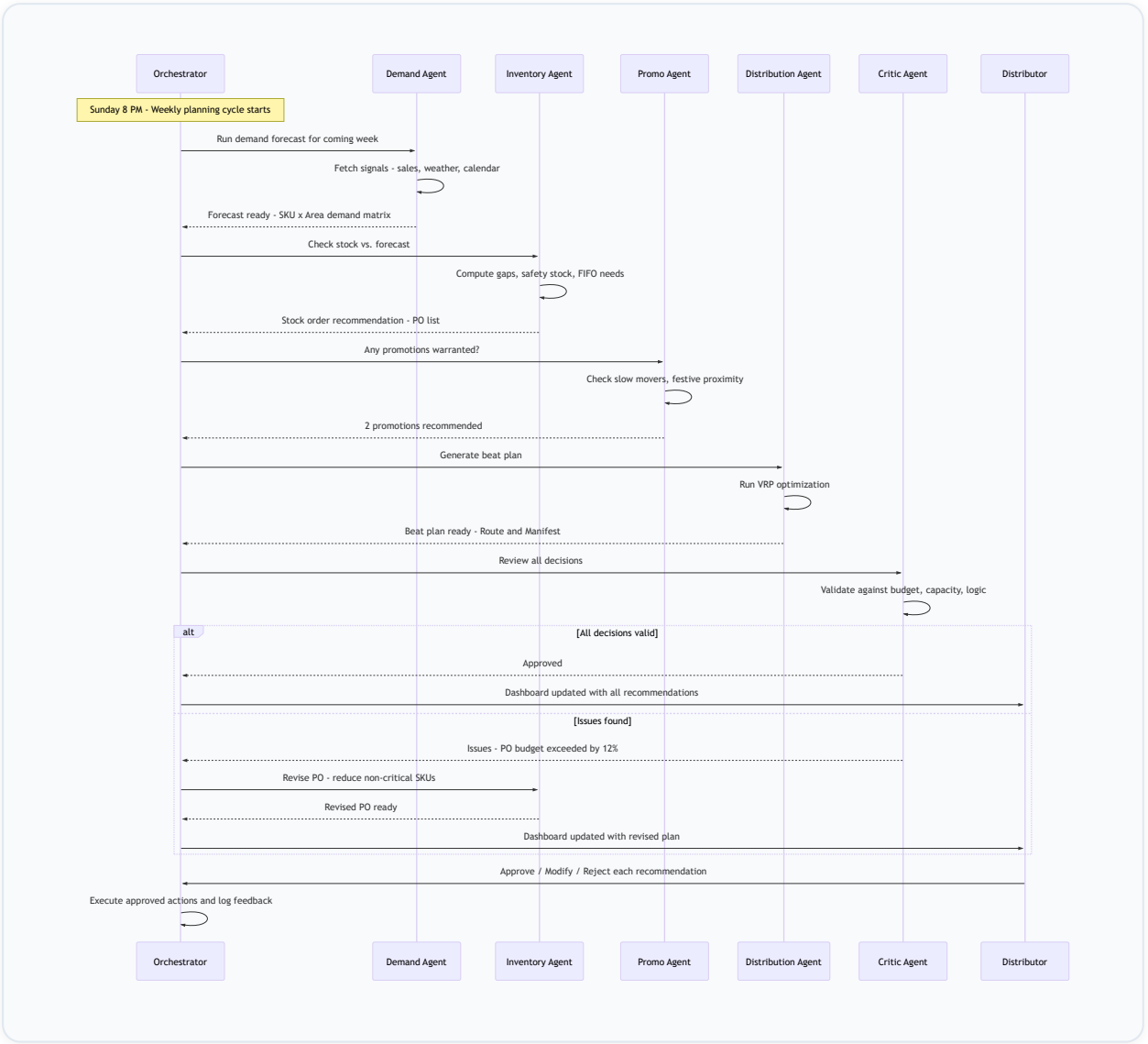


7.6 Feature 6: Quick Commerce Integration

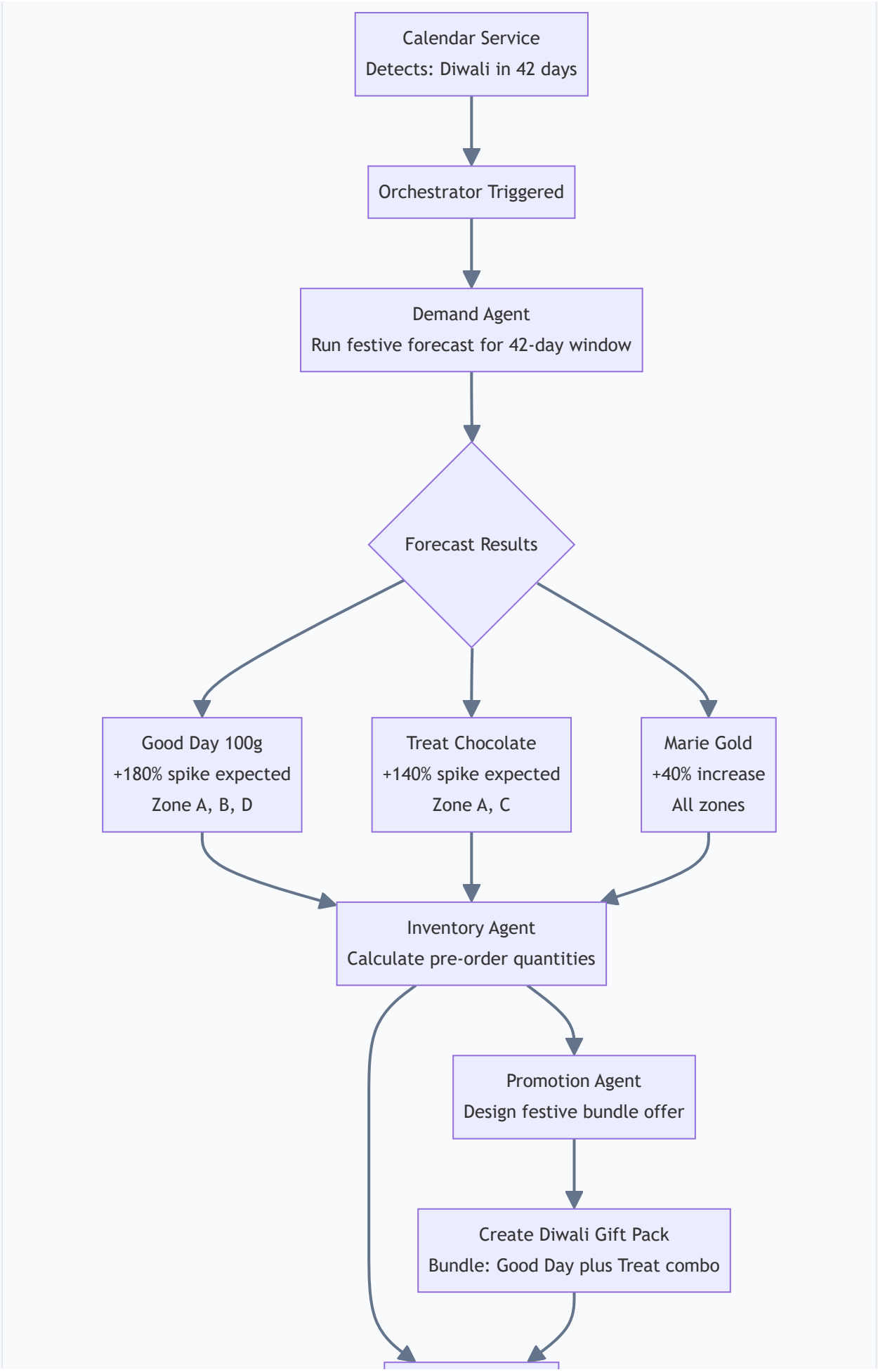


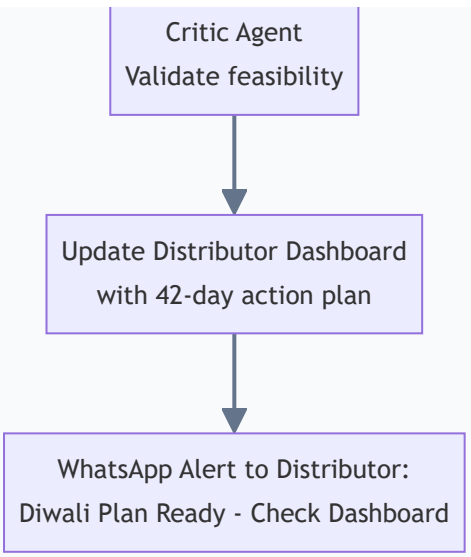
8. Agent Interaction Diagrams

8.1 Weekly Planning Cycle — Full Agent Interaction



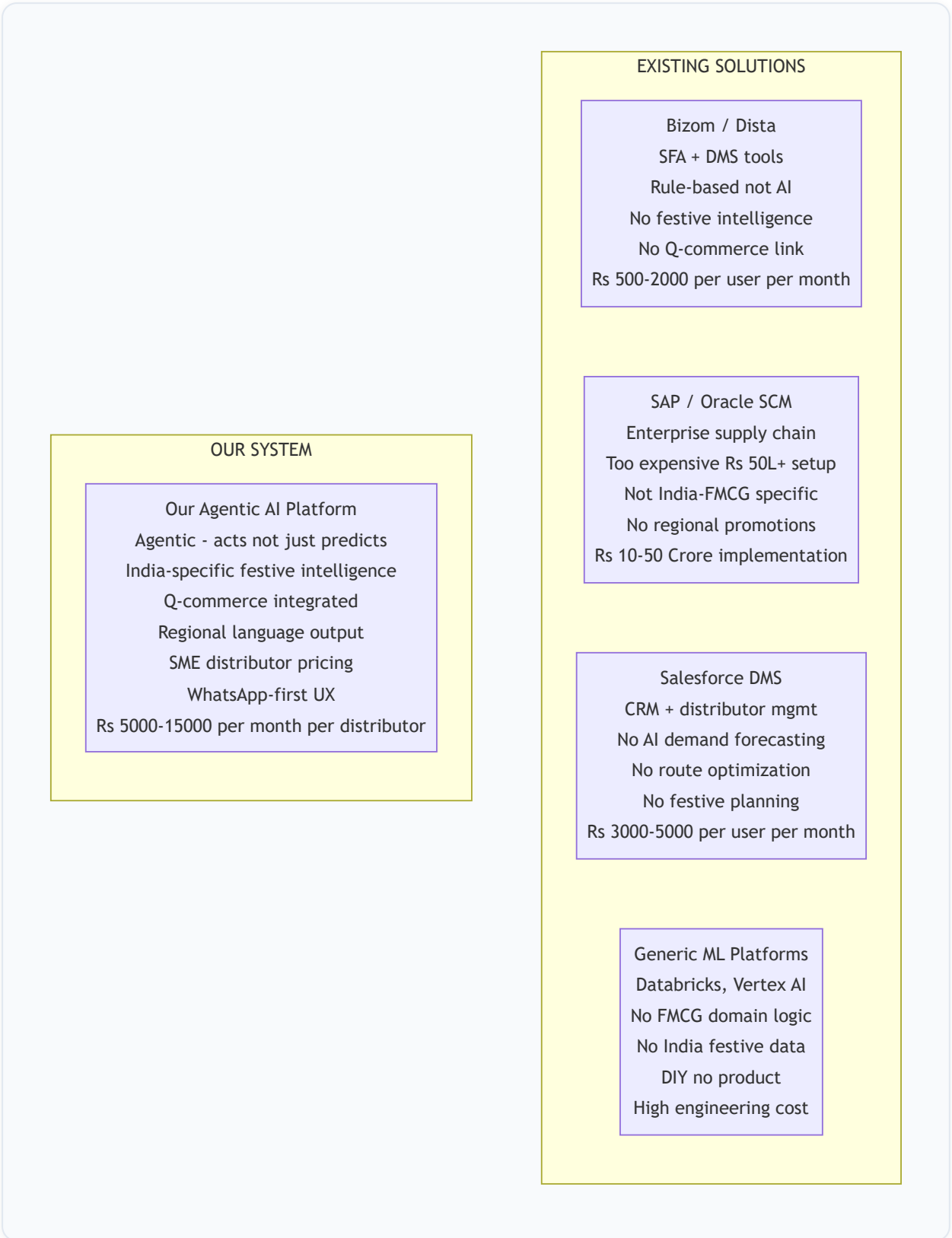
8.2 Festive Event Detection & Response


Calendar Service
Detects: Diwali in 42 days
Orchestrator Triggered
Demand Agent
Run festive forecast for 42-day window
Forecast Results
Good Day 100g
+180% spike expected
Zone A, B, D
Treat Chocolate
+140% spike expected
Zone A, C
Marie Gold
+40% increase
All zones
Inventory Agent
Calculate pre-order quantities
Promotion Agent
Design festive bundle offer
Create Diwali Gift Pack
Bundle: Good Day plus Treat combo


Critic Agent
Validate feasibility
Update Distributor Dashboard
with 42-day action plan
WhatsApp Alert to Distributor:
Diwali Plan Ready - Check Dashboard

9. Gap Analysis — What Exists vs. What We Build

9.1 Competitive Landscape

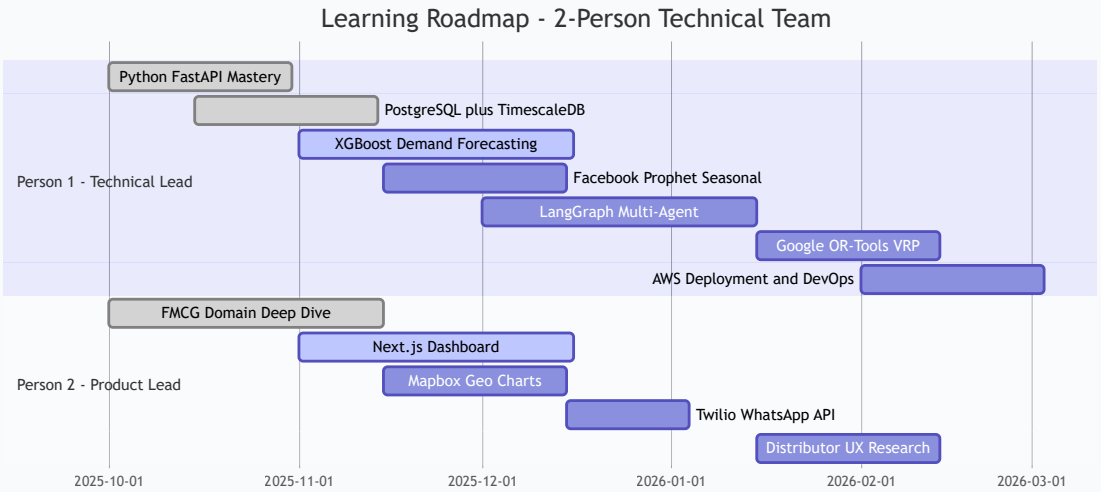


9.2 Feature Gap Matrix

FEATURE	BIZOM	SAP	SALESFORCE	OUR SYSTEM
Area × SKU demand heatmap	⚠️ Basic	✅ Yes	❌ No	✅ **AI-powered**
Agentic auto-ordering	❌ No	⚠️ Rule-based	❌ No	✅ **AI agent**
Festive demand prediction	❌ No	⚠️ Manual	❌ No	✅ **AI + calendar**
Regional language offers	❌ No	❌ No	❌ No	✅ **LLM-generated**
Q-commerce integration	❌ No	⚠️ Custom	❌ No	✅ **Built-in**
WhatsApp distributor bot	❌ No	❌ No	❌ No	✅ **Phase 5**
FIFO batch management	⚠️ Basic	✅ Yes	❌ No	✅ **AI agent**
SME pricing under ₹15K/month	✅ Yes	❌ No	❌ No	✅ **Yes**
Continuous learning	❌ No	❌ No	❌ No	✅ **Feedback loop**

10. What We Need to Learn

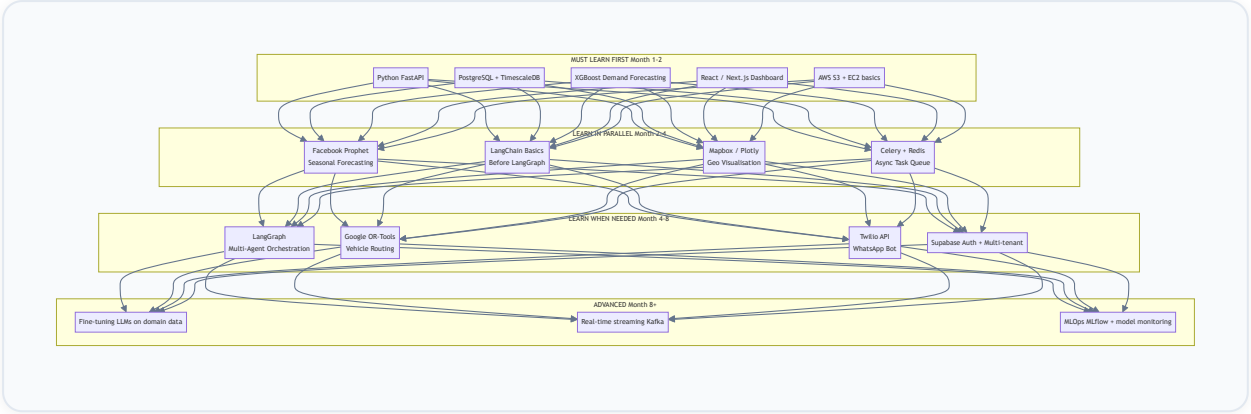
10.1 Technical Learning Roadmap



10.2 Domain Knowledge We Must Acquire

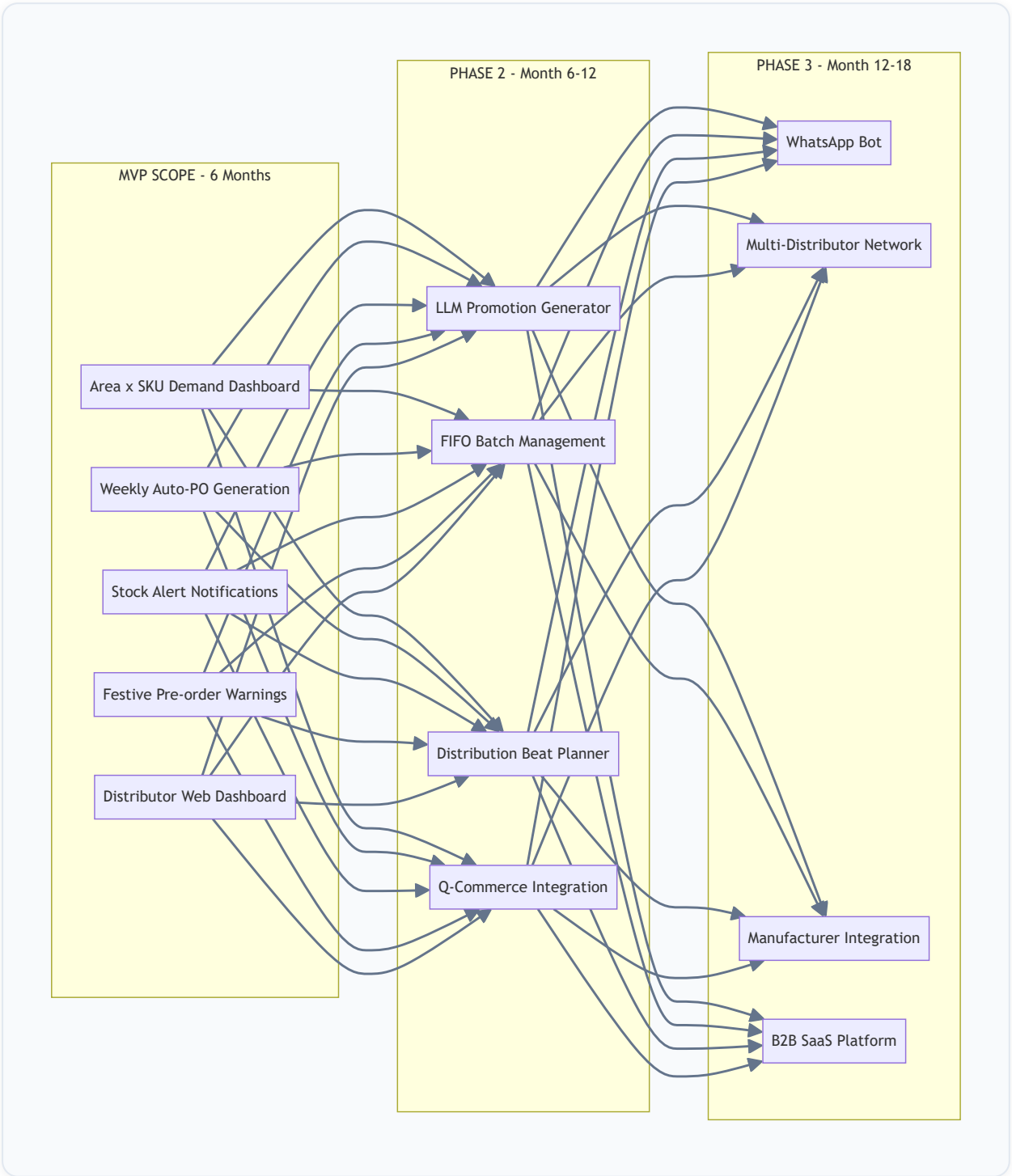
AREA	WHAT TO LEARN	HOW TO LEARN	PRIORITY
FMCG distribution mechanics	Trade margins, credit cycles, scheme structures	Shadow 2–3 distributors for 1 week each	🔴 Critical
Biscuit SKU lifecycle	Shelf life, FIFO, expiry policies	Distributor warehouse visits	🔴 Critical
India festive calendar nuances	State-by-state festival timing	Regional FMCG sales manager interviews	🟡 High
Tally/DMS integration	API formats, data schemas	Tally developer documentation	🟡 High
Q-commerce dark store ops	Blinkit/Swiggy ops workflow	Partnership conversations	🟠 Medium
GST data access legally	e-invoicing APIs, consent frameworks	GSTN developer portal	🟠 Medium

10.3 Technology Learning Priority Path

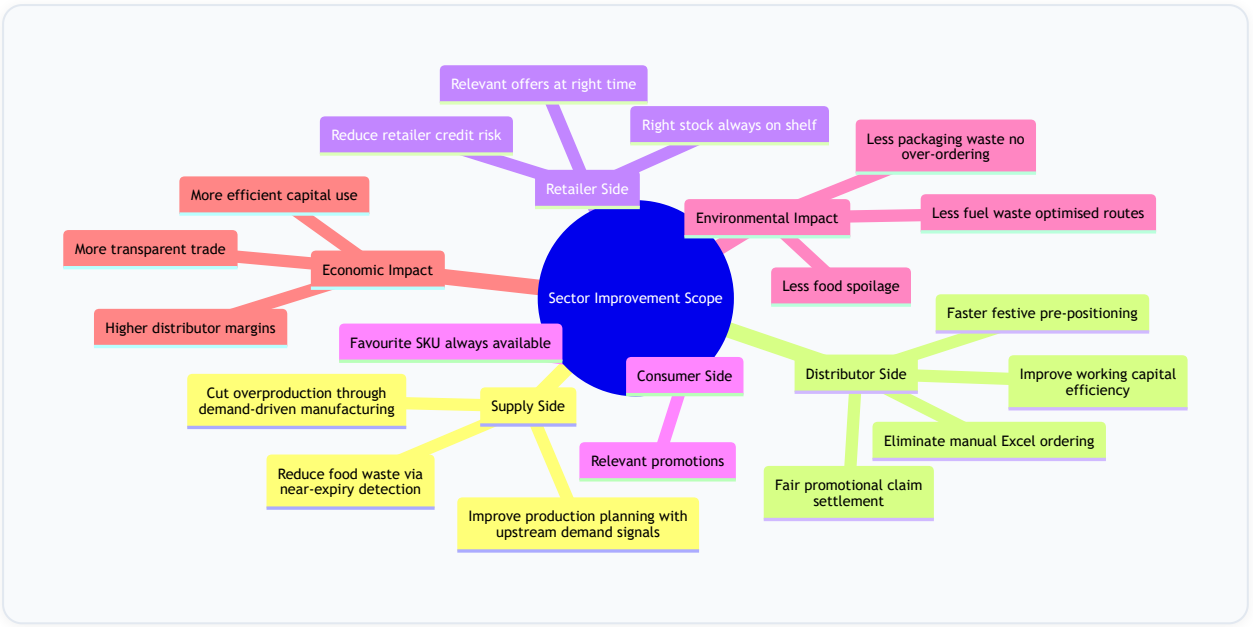


11. Scope to Build & Improve

11.1 MVP vs. Full Product Scope



11.2 Sector Improvement Scope — Beyond This System

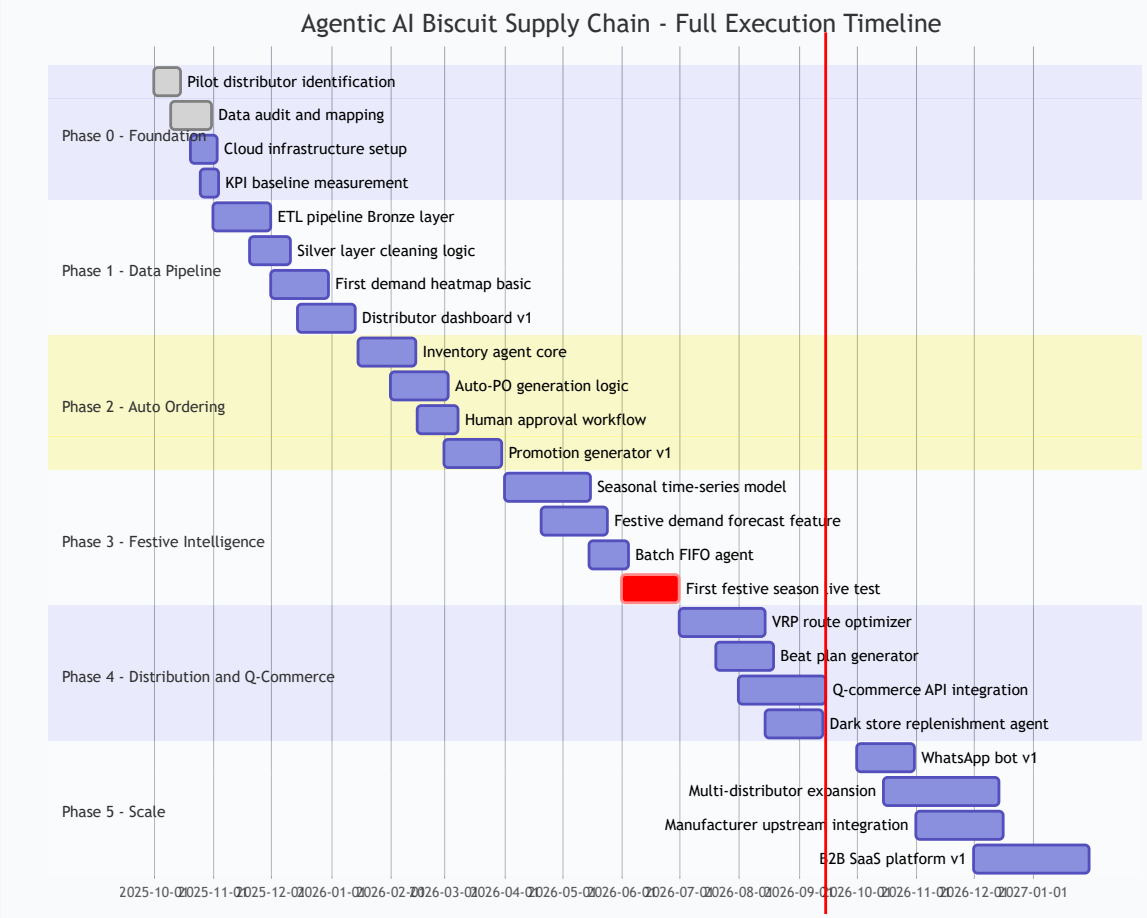


11.3 Future AI Capabilities — Research Frontier

CAPABILITY	RESEARCH STAGE	IMPLEMENTATION TIMELINE
Computer vision (shelf monitoring via retailer phone camera)	Applied research	24–36 months
LLM negotiation agent (auto-negotiate with supplier on price)	Emerging	18–24 months
Generative demand simulation (synthetic data for rare events)	Research	12–18 months
Federated learning (learn across distributors without sharing raw data)	Applied research	24–36 months
Voice-first distributor interface (speak to system in Hindi)	Ready today	12–18 months
Carbon footprint tracking per delivery route	Available	18–24 months

12. Phase-wise Execution Plan

12.1 Master Timeline — Gantt View



12.2 Phase-by-Phase Key Tasks

PHASE 0: Foundation & Discovery (Weeks 1–6)

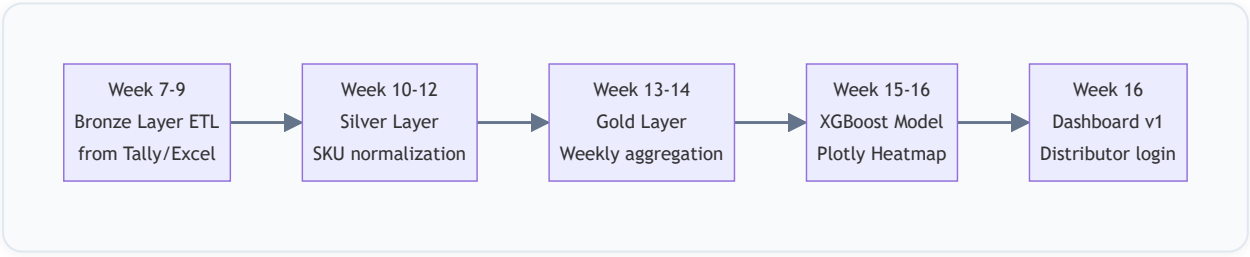
TASK	WHO	OUTPUT
Visit 3–5 distributors, shadow daily workflow	Person 2	Process documentation
Collect 12–24 months past sales data	Person 2	Raw CSV/Excel files
Data quality audit	Person 1	Data quality report
Set up AWS (EC2, RDS, S3)	Person 1	Cloud environment
Define 5 KPIs with baseline measurements	Both	KPI tracking doc
Legal: NDA with pilot distributor	Both	Signed NDA

KPIs to baseline:

1. Current stockout rate (%)
2. Current forecast accuracy (if any)
3. Time taken to prepare weekly stock order (hours)
4. Festive stockout rate last year
5. Delivery efficiency (stops per day per vehicle)

PHASE 1: Data Pipeline & First Intelligence (Weeks 7–16)

Milestone: Distributor can see for the first time — "which of my areas is selling what."



PHASE 2: Auto-Ordering & Promotions (Weeks 17–26)

Milestone: "Distributor clicks Approve on an AI-generated PO for the first time."

Key deliverables:

- Inventory Agent (stock gaps, safety stock, order qty per SKU)
- Draft PO generation UI with line-by-line review
- One-click approve workflow
- LLM-based promotion generator (LangChain + GPT-4/Gemini API)
- Basic SMS notification for distributor

PHASE 3: Festive Intelligence (Weeks 27–36)

Critical success test: Compare stockout rate during Diwali 2026 vs. Diwali 2025 (manual). Target: >40% reduction.

Key deliverables:

- Prophet-based seasonal model with festival features
- 6–8 week ahead festive forecast dashboard view
- Calendar widget showing demand intensity by week
- FIFO batch tracking + expiry alert agent
- Batch pre-order workflow (raise POs 6–8 weeks early)

PHASE 4: Distribution & Q-Commerce (Weeks 37–48)

Key deliverables:

- Google OR-Tools VRP integration
- Beat plan generator with map view
- Retailer priority scoring (high-velocity retailers first)
- Q-commerce dark store API integration
- Real-time dark store inventory webhook listener
- Auto-trigger replenishment for dark store stockouts

PHASE 5: Scale (Months 13–18)

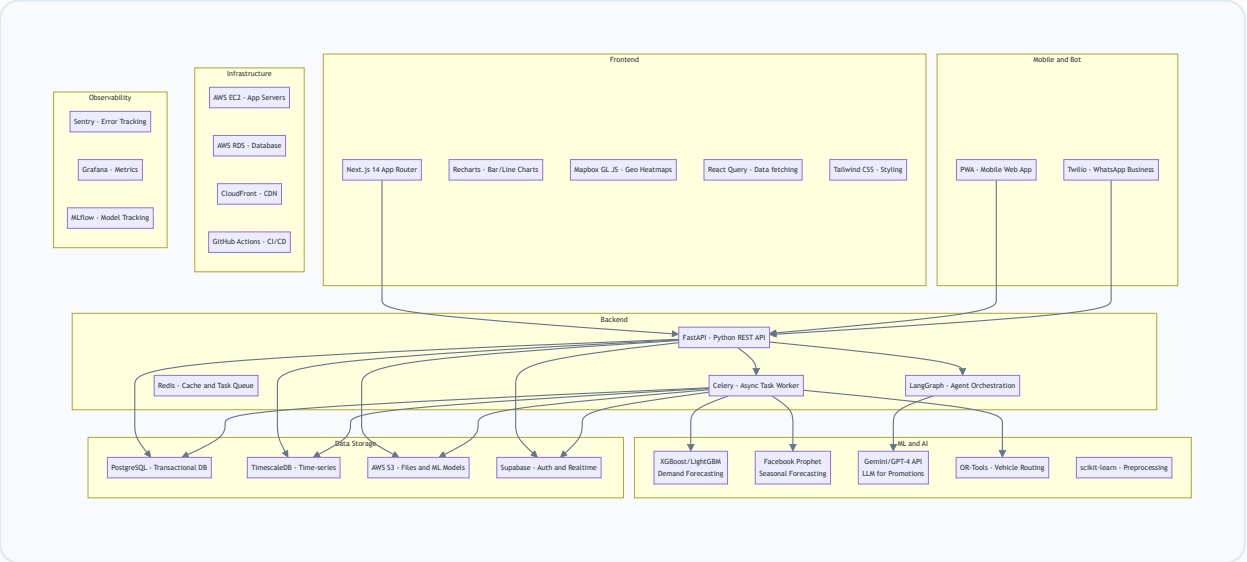
Key deliverables:

- WhatsApp bot (Twilio) — distributor asks in Hindi: "How much Good Day should I order?"
- Multi-distributor network (5+ distributors on platform)
- Manufacturer integration: anonymized demand signals fed upstream
- B2B SaaS packaging: pricing tiers, self-serve onboarding
- Competitor price intelligence (web scraping + alert)

---

13. Technology Stack — Deep Dive

13.1 Full Stack Map

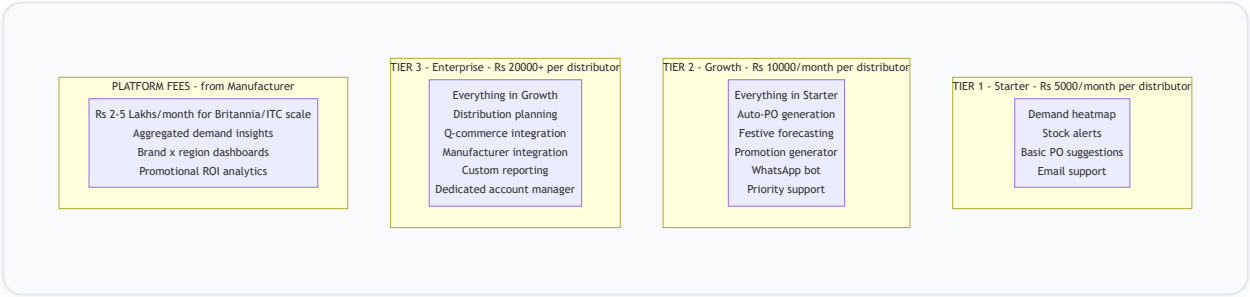


13.2 Monthly Cloud & API Cost Estimate

SERVICE	MVP STAGE	GROWTH STAGE	SCALE STAGE
AWS EC2 (t3.medium)	Free tier	~$35/month	~$150/month
AWS RDS (PostgreSQL)	Free tier	~$25/month	~$100/month
AWS S3	5GB free	~$5/month	~$20/month
OpenAI/Gemini API	$5 credits	~$50/month	~$300/month
Mapbox (geo maps)	50K loads free	~$20/month	~$100/month
Twilio WhatsApp	Pay per message	~$30/month	~$150/month
TOTAL	**~$0**	**~$165/month**	**~$820/month**

14. Business Model & Monetisation

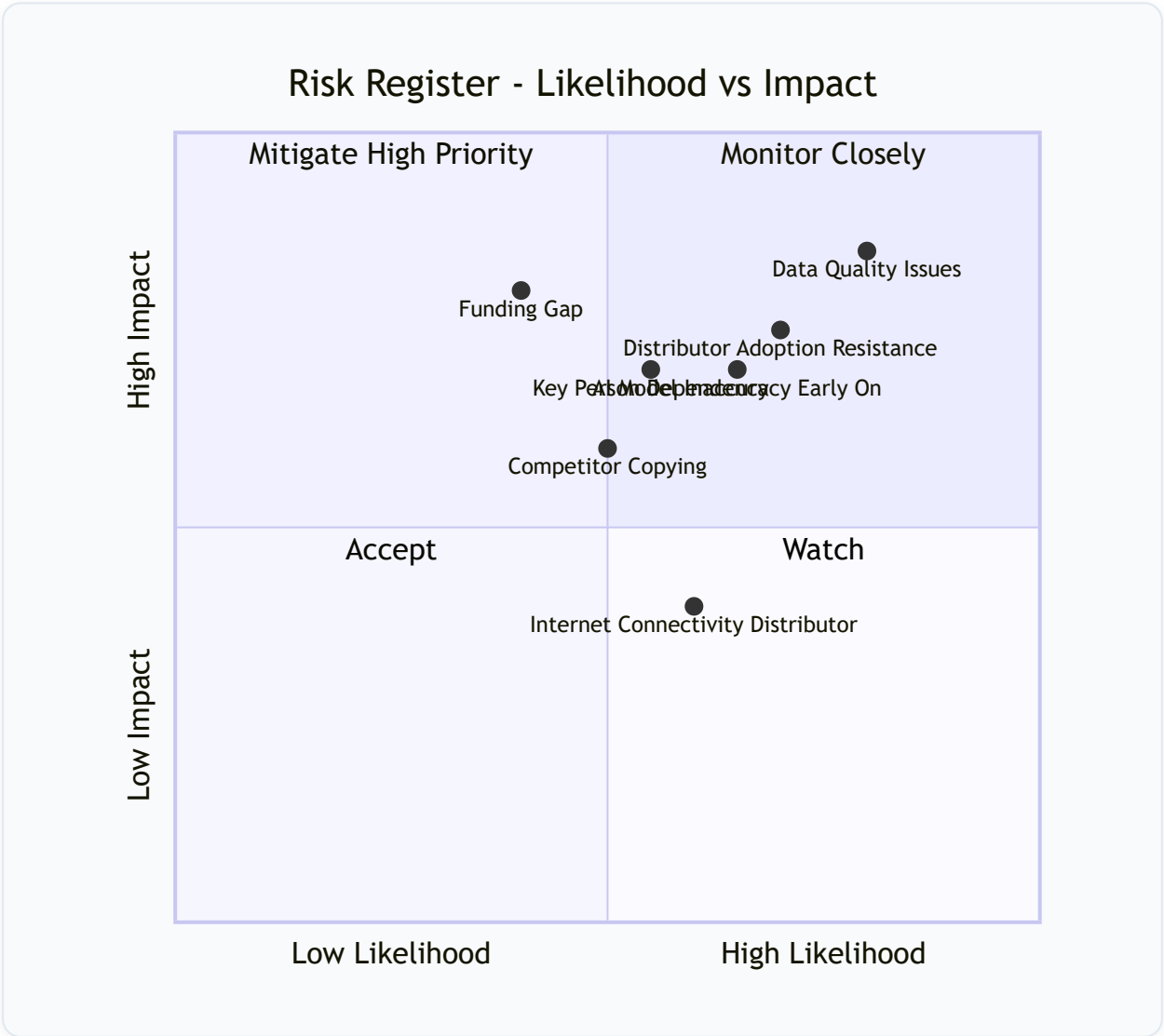
14.1 Revenue Tiers



14.2 Unit Economics Projection

METRIC	YEAR 1	YEAR 2	YEAR 3
Distributors on platform	10	80	300
Average revenue per distributor	₹8,000/month	₹9,000/month	₹10,000/month
Monthly Recurring Revenue	₹80,000	₹7.2 Lakhs	₹30 Lakhs
Annual Recurring Revenue	₹9.6 Lakhs	₹86 Lakhs	₹3.6 Crore
Gross margin (target)	65%	72%	78%

15. Risk Register



RISK	MITIGATION
Data quality is poor	Start with 1 clean distributor dataset; build validation into ETL from day 1
Distributors won't adopt	Give them value FIRST (heatmap dashboard free); charge only after proven value
AI model inaccuracy early	Show human-reviewable recommendations; never auto-execute without approval initially
Competitor copies	Build network effects: more distributors = better model = harder to replicate
Funding gap	Bootstrap Phase 0–2; seek first revenue before Phase 3; apply for grants
Key person dependency	Document everything; cross-train both team members on critical paths

TIP

> *Apply for* ***STARTUP INDIA*** *,* ***NASSCOM DEEPTECH*** *, or* ***GOOGLE FOR STARTUPS*** *grants — AI-powered supply chain tools qualify under multiple government funding schemes.*

16. References & Sources

#	SOURCE	TYPE	YEAR
1	Deloitte: "Agentic AI in Supply Chains"	Industry Report	2024
2	IBM Institute for Business Value: "AI in FMCG"	Research	2024
3	McKinsey Operations Practice: "Supply Chain Reinvention"	Consulting	2023
4	Kanerika: "AI-Driven FMCG Case Studies"	Industry	2024
5	Accenture Supply Chain Survey	Research	2024
6	IIT Kanpur: "AI Applications in Indian FMCG"	Academic	2023
7	arXiv: "Multi-Agent Systems for Supply Chain"	Academic	2024
8	NTT Data: "Britannia Supply Chain Optimization"	Case Study	2023
9	Analytics India Mag: "Britannia Digital Transformation"	Media	2023
10	Redseer / Bernstein: "India Q-Commerce Market Report"	Market Research	2024
11	LangGraph Documentation: "Multi-Agent Architecture"	Technical	2024
12	CrewAI Documentation: "Agentic Workflows"	Technical	2024
13	Google OR-Tools: "Vehicle Routing Problem"	Technical	2024
14	NASSCOM FMCG Tech Report	Industry	2024
15	SAP: "Agentic AI in Supply Chain Networks"	Technical	2024

© 2025 — Agentic AI Biscuit Supply Chain Project

This document is a living reference — update as research and implementation progress.

Version 1.0 | Document created: September 2025